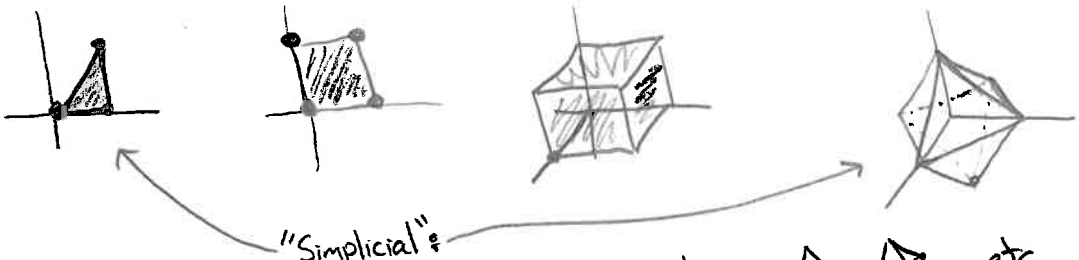
First Class:

-Def. A polytope, P, is the convex hull of a set of vertices in $\mathbb{R}^N$. ie. $v_1, \dots, v_k \in \mathbb{R}^N \Rightarrow \left\{ \sum_{i=1}^{k} \lambda_i v_i \;\middle|\; 0 \le \lambda_i \le 1 \right\}$

-Exs:



"Simplicial": all proper facets are simplexes $\triangle$, $\triangle$, etc.

Face vector: $f = (f_0, f_1, \dots, f_{d-1})$, f_i = # of i-dimensional faces

Q: Given a vector f, when is there an associated polytope? [Simpler: can we answer for what f there is an associated simplicial polytope?]

McMullen: Better to associate an h-vector to P.

→ Inductive definition (induction on faces of P)

→ For simplicial P, $h(P) = (h_0, \dots, h_d)$ are coeffs. of

$$h(P,t) = (t-1)^d + f_0(t-1)^{d-1} + \cdots + f_{d-1}$$

e.g. $P = \Delta_n$ a n-diml simplex

$\Rightarrow h(\Delta_n, t) = t^d + t^{d-1} + \cdots + 1$

P = cube $\Rightarrow h(P,t) = t^3 + 5t^2 + 5t + 1$

Conjecture (McMullen) / Thm (Stanley): $h \in \mathbb{Z}_{\ge 0}^{d+1}$ is an h-vector for a d-diml simplicial polytope P iff

① $h_i = h_{d-i}$ $\forall i$

② $\exists$ graded $\mathbb{Q}$-alg. R with $R_0 = \mathbb{Q}$, R generated in deg 1, $\dim_{\mathbb{Q}} R_i = h_i - h_{i-1}$ $\forall 1 \le i \le \lfloor d/2 \rfloor$

Cor. P: simplicial $\Rightarrow$ $h(P)$ is unimodal

Stanley's idea: $P \longleftrightarrow$ "fans" $\longleftrightarrow$ toric variety $X_P / \mathbb{C}$

Thm: P: simplicial $\Rightarrow$ $h(P,t) = \sum_{k=0}^{d} \dim H^{2k}(X_P, \mathbb{Q}) \cdot t^k$ & $H^{odd}(X_P, \mathbb{Q}) = 0$

When P is not simplicial, X_P is not "$\mathbb{Q}$-smooth" & Stanley's theorem fails. But there is a fix!

② **Thm (Stanley):** For general rational P,

$$h(P,t) = \sum_{k=0}^{d} \dim \underbrace{IH^{2k}(X_P, \mathbb{Q})}_{\text{"Intersection homology"}} \cdot t^k$$

~ We will return later in the course to prove this. ~

What is "Intersection (co)homology"?

A: provides a good (co)homology theory for singular varieties

"good": we can do intersection theory with it.

Intersection Theory for smooth varieties:

X: smooth, oriented n-manifold, $H_n(X;\mathbb{Z}) \cong \mathbb{Z} \ni [X]$ fundamental class

Thm (Poincaré Duality). $- \cap [X] : H^i(X,\mathbb{Z}) \longrightarrow H_{n-i}(X,\mathbb{Z})$

So with rational coeffs:

$$\begin{array}{ccc} H^i(X,\mathbb{Q}) & \xrightarrow{\sim} & H_{n-i}(X,\mathbb{Q}) \\ \| & \text{universal coeffs. thm} & \| \\ H_i(X,\mathbb{Q})^* & & H^{n-i}(X,\mathbb{Q})^* \end{array}$$

Cor: $h_i = \dim_{\mathbb{Q}} H_i(X,\mathbb{Q}) = \dim_{\mathbb{Q}} H^{n-i}(X,\mathbb{Q})$ is a symmetric/palandromic sequence.

Better: can specify the isomorphism $H^{n-i}(X,\mathbb{Q}) \cong H^{n+i}(X,\mathbb{Q})$

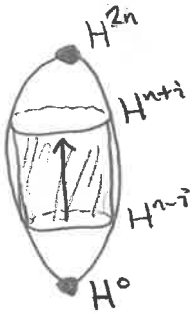
Assume $X \subseteq \mathbb{P}^N_{\mathbb{C}}$ is a complex, projective variety.

Choose a generic hyperplane $H \subset \mathbb{P}^N$

$\rightsquigarrow [X \cap H] \in H_{2n-2}(X,\mathbb{Z}) \overset{\text{P.D.}}{\Longleftrightarrow} [H] \in H^2(X,\mathbb{Z})$

Thm (Hard Lefschetz). $H^{n-i}(X,\mathbb{Q}) \xrightarrow[\sim]{-\cup [H]^i} H^{n+i}(X,\mathbb{Q})$ is isom.

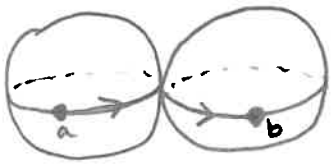
↳ Rmk on how "hard" Hard Lefschetz is: hard enough Lefschetz didn't prove it in his 1924 paper; Hodge did in 1941!



Cor. h_i are also unimodal.

③ This fails for singular spaces

Ex: $X = \mathbb{P}^1 \cup \mathbb{P}^1 \subseteq \mathbb{P}^2$



By Mayer-Vietoris:

$$H_i(X,\mathbb{Q}) = \begin{cases} \mathbb{Q} & i=0 \\ 0 & i=1 \\ \mathbb{Q}\oplus\mathbb{Q} & i=2 \end{cases}$$

no Poincaré duality

$H^i(X,\mathbb{Q})^*$

What is going on? a & b want to be distinct 0-cycles, but they are connected by a 1-cocycle through the singularity.

IH will exclude this cocycle:

$$IH^i(X,\mathbb{Q})^* \cong IH_i(X,\mathbb{Q}) = \begin{cases} \mathbb{Q}\oplus\mathbb{Q} & i=0 \\ 0 & i=1 \\ \mathbb{Q}\oplus\mathbb{Q} & i=2 \end{cases}$$

But what if I only care about cohomology? IH has a sheafy analogue called perverse sheaves.

cohomology sheaf of a proper map $\in$ { perverse sheaves }

The category of perverse sheaves has atoms called "Intersection Cohomology" (IC) sheaves.

Thm [Decomposition Thm]. $f: X \to B$ a proper map between smooth varieties, then $Rf_* \overline{\mathbb{Q}}_\ell$ is a semisimple perverse sheaf, i.e. it is a direct sum of IC sheaves.

$\Longrightarrow$ If you know the supports of summands, you can go from general fibers to smooth fibers!

Underlies Springer theory, proof of Fundamental Lemma, Kazhdan-Luzstig theory, proof of P=W conjecture, ... many many more!

(3.5)

Recap of some algebraic topology:

Chains: Δ^n = n-simplex

$C_n(X)$ = free abelian group generated by maps $\Delta^n \longrightarrow X$

$$\partial_n(\sigma) = \sum_{i=0}^{n} (-1)^i \cdot (\sigma \circ d_i)$$ for $d_i : \Delta^{n-1} \hookrightarrow \Delta^n$ the ith face.

$$H_i(X) = H_i(\cdots \to C_n(X) \to C_{n-1}(X) \to \cdots)$$

Rmk: Can replace $C_n(X)$ with set of (possibly infinite) sums $\sum c_i \sigma_i$ s.t. every cpt $K \subset X$ intersects only finitely many σ_i. In this case, the complex computes Borel-Moore homology, a sort of "homology w/cpt supports".

e.g. $\sigma_i : \Delta^1 \longrightarrow \mathbb{R}$ by the interval $[i, i+1]$

$$\sum_{i \in \mathbb{Z}} \sigma_i \in C_1^{BM}(\mathbb{R})$$

Exercise: $\sum_{i \in \mathbb{Z}} \sigma_i$ generates $H_1^{BM}(\mathbb{R}) \cong \mathbb{Z}$.

Whitney Stratifications (CORRECTED): $n=\dim_{\mathbb{R}}(X)$

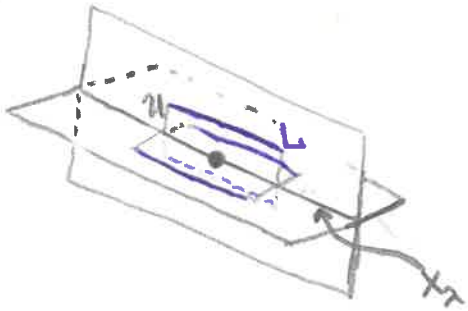
$X=\bigsqcup_\lambda X_\lambda$ is a stratification if $\overline{X_\lambda}$ is a union of X_μ for some set of μ.

This stratification is a Whitney stratification if

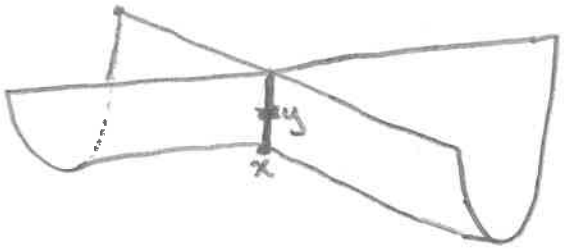
$\forall \lambda$, there is a stratified space $L=\bigsqcup_\mu L_\mu$ of $\dim(L)=n-\dim(X_\lambda)-1$ s.t. $\forall x\in X_\lambda$, $\exists$ neighborhood $\mathcal{U}\ni x$ s.t. $\phi: \mathcal{U} \overset{\text{homeo.}}{\approx} \mathbb{R}^{\dim(X_\lambda)} \times \operatorname{cone}(L)$, where

$$\operatorname{cone}(L)= L\times[0,1)/L\times\{0\}$$

s.t. $\phi^{-1}(L_\mu)=X_\mu\cap\mathcal{U}$ & $\phi|_{\mathcal{U}\cap X_\lambda}: \mathcal{U}\cap X_\lambda \xrightarrow{\sim} \mathbb{R}^{\dim(X_\lambda)}\times\{0\}$

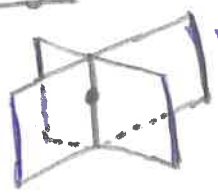


Ex: Whitney Umbrella: $\{x^2=zy^2\}\subseteq\mathbb{C}^3$



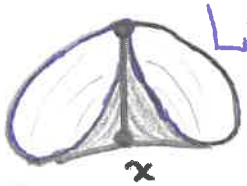
At y:

$y\in U$:



$\approx \mathbb{R}\times\operatorname{cone}(L)$

At x:



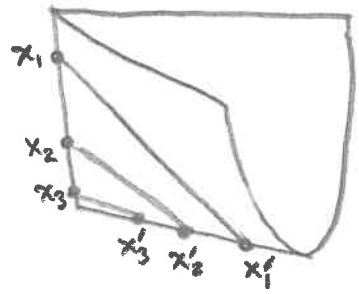
$L=S^1\vee S^1=(S^1\vee S^1\setminus\{pt\})\sqcup\{pt\}$ (for real picture)

These are equivalent to the Whitney conditions:

(a) If $x_i \longrightarrow x$ for $x_i \in X_\lambda$ & $x \in X_\mu$, then $T_x X_\mu \subseteq \lim_{i \to \infty} T_{x_i} X_\lambda$ if this limit exists

(b) If $x_i \in X_\lambda$ & $x_i' \in X_\mu$ both go to $x \in X_\lambda$, then $\lim_{i \to \infty} [\text{Secant line } (x_i, x_i')] \subseteq \lim_{i \to \infty} T_{x_i} X_\lambda$ if limits exist.

Ex: for Whitney umbrella, can take



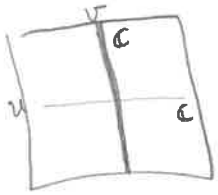
limit of secant lines is horizontal, but $\lim_{i \to \infty} T_{x_i} X_\lambda$ is vertical line only

So we need extra strata.

(Complex) Link of Whitney Umbrella:

$\pi: \mathbb{C}^2 \to X$
$(u,v) \mapsto (uv, v, u^2)$

compatible w/ contraction $\lambda \cdot (u,v) = (\lambda u, \lambda^2 v)$ & $\lambda \cdot (x,y,z) = (\lambda^3 x, \lambda^2 y, \lambda^2 z)$



$\longrightarrow$ glues $\pm u$ coords

Link at $(0,0)$ is $X \setminus \{0\} / G_n \cong \mathbb{C}^2 \setminus \{0\} / \big((u,0) = (-u,0)\ \forall u,\ \lambda(\lambda u, v) = (u,v)\ \forall \lambda \forall u,v\big)$

$\cong S^3 / \big((u,0) = (-u,0)$ on an unknotted $S^1 \subset S^3\big)$

Links of Isolated Singularities:

Thm (Milnor fibers). $f: \mathbb{C}^{n+1} \to \mathbb{C}$ s.t. $f(0) = 0$, f: holomorphic

Let $B_\varepsilon = \{z \in \mathbb{C}^{n+1} \mid \|z\| \le \varepsilon\}$ be a small ball @ $0 \in \mathbb{C}^{n+1}$.

Then $f^{-1}(0) \cap B_\varepsilon \underset{\text{homeo.}}{\approx} \text{cone}(L)$ for $L \underset{\text{homotopic}}{\approx} \bigvee_\mu S^n$.

for $\mu = \dim_{\mathbb{C}} \dfrac{\mathbb{C}[\![x_0, \dots, x_n]\!]}{\left(\frac{\partial f}{\partial x_0}, \dots, \frac{\partial f}{\partial x_n}\right)}$

Ex: $f: \mathbb{C}^2 \to \mathbb{C}$, $(x,y) \mapsto y^2 - x^3$ has $\mu = \dim_{\mathbb{C}} \dfrac{\mathbb{C}[\![x,y]\!]}{(2y, -3x^2)} = \dim_{\mathbb{C}} \text{Span}\{1, x\} = 2$

$L \approx S^1 \vee S^1$

Complex Link of $X = \{y^2 = x^3\} \subseteq \mathbb{C}^2$:

$\pi: \mathbb{C} \to X$, $z \mapsto (z^2, z^3)$

Write $z = re^{i\theta}$ & fix r.

$\Rightarrow (r^2 e^{2i\theta}, r^3 e^{3i\theta})$ traverses 1 circle 3 x's & the other twice.

$\Rightarrow$ torus knot in S^3.

LECTURE 2:

Intersection Homology: X: singular $\mathbb{C}$-variety.

0: Sings. are interesting. e.g. $x^n = y^m$

I: Whitney Stratifications

$X = \coprod_\lambda X_\lambda$ stratification (ie. X_λ connd, locally clsd, smooth; $\overline{X_\lambda} = \bigcup_\mu X_\mu$)

We ask X_λ is "equisingular", i.e. $\forall x \in X_\lambda$, $\exists$ nghd $U \ni x$ s.t. $(X_\lambda \cap U) \times V \cong U$ (id on $X_\lambda \cap U$) ("local normal triviality")

Ex: Whitney umbrella $\{x^2 = zy^2\} \subseteq \mathbb{C}^3$.

$\mathbb{C}^2 \to$  $X_{reg} \cup X_{sing}$ not good enough!

$(u,v) \mapsto (uv, v, u^2)$ $X_{reg} \cup (X_{sing} \setminus 0) \cup \{0\}$

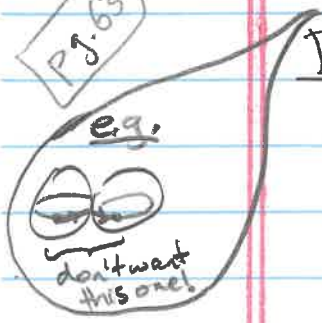
$y^2 + x^3 - t^2 x^2 = 0$: 

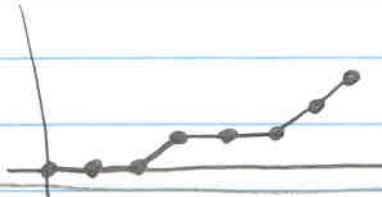
Whitney Condition: $X_\mu \subset \overline{X_\lambda} \Rightarrow \forall a_i \in X_\lambda$, $b_i \in X_\mu$ converging to $c \in X_\mu \to \lim_{i\to\infty}$ (secants $a_i - b_i$) $\in \lim_{i\to\infty} T_{a_i} X$

Intersection Homology: to recover P.D. / intersection thy, need to restrict what cycles we deal with.

Def: A perversity is a fxn $\bar{p} : [0, \dim X] \cap \mathbb{Z} \to \mathbb{N}$ (real dimension)

s.t. $p_\alpha := \bar{p}(\alpha)$ satisfies $\begin{cases} p_0 = p_1 = p_2 = 0 \\ p_\alpha \le p_{\alpha+1} \le p_\alpha + 1, & \alpha \ge 2 \end{cases}$

p. 63 e.g.  don't want this one!

 slopes 0 or 1 everywhere.

e.g. upper middle perversity: $(0,0,0,1,1,2,2,\dots,\frac{n}{2}-1)$ ★ DEFAULT

lower middle: $(0,0,0,0,1,1,2,2,\dots,\frac{n}{2}-1,\frac{n}{2}-1)$

$X = X_n \supset X_{n-1} \supset \cdots \supset X_1 \supset X_0 \supset X_{-1} = \emptyset$ $(\dim X_i \le i)$

An i-chain $\xi \in C_i(X)$ is $\bar{p}$-allowed if $\dim(|\xi| \cap X_{n-\alpha}) \le i - \alpha + p_\alpha$

Note: $\xi \pitchfork X_{n-\alpha} \iff \dim(|\xi| \cap X_{n-\alpha}) = i - \alpha$

$$IC_i^{\bar{p}}(X) = \left\{ \xi \in C_i(X) \;\middle|\; \begin{array}{l} \dim(|\xi| \cap X_{n-\alpha}) \le i - \alpha + p_\alpha \\ \dim(|\partial\xi| \cap X_{n-\alpha}) \le i - 1 - \alpha + p_\alpha \end{array} \forall \alpha \ge 2 \right\}$$

$$IH_i^{\bar{p}}(X;\mathbb{Z}) = H_i\left(IC_\bullet^{\bar{p}}(X)\right)$$

⑤ • Ex: $X = \mathbb{P}^1 \cup \mathbb{P}^1 \subseteq \mathbb{P}^2$, $X_2 = \mathbb{A}^1 \cup \mathbb{A}^1 \supseteq X_1 = X_0 = \{pt\}$

$p_0 = p_1 = p_2 = 0$ is only option for perversity fxn.

$$IC_i(X) = \left\{ \xi \in C_i(X) \;\middle|\; \begin{array}{l} \dim(|\xi| \cap X_0) \leq i-2 \\ \dim(|\partial\xi| \cap X_0) \leq i-3 \end{array} \right\}$$

Chains can only touch singularity at $i=2$ (impossible due to this)

Chain boundaries can never touch the singularity

$$\Rightarrow IC_\bullet(X) \simeq C_\bullet(\mathbb{P}^1 \sqcup \mathbb{P}^1)$$

$$\& \quad IH_i(X) = H_i(\mathbb{P}^1 \sqcup \mathbb{P}^1) = \begin{cases} \mathbb{Q} \oplus \mathbb{Q} & i = 0, 2 \\ 0 & i = 1. \end{cases}$$

More generally:

• Ex: X: $2m$-dimensional with an isolated singular point x

$$X = X_{2m} \supset X_0 = \{x\}.$$

Let's use the middle perversity

Prop: $$IH_i(X, \mathbb{Z}) = \begin{cases} H_i(X, \mathbb{Z}), & i > m \\ \mathrm{Im}(H_m(X-x, \mathbb{Z}) \to H_m(X, \mathbb{Z})), & i = m \\ H_i(X-x, \mathbb{Z}), & i < m \end{cases}$$

Pf: $\xi \in IC_i^{\underline{m}}(X)$ iff ① $\dim(|\xi| \cap X_0) \leq i - 2m - \underline{m}(2m) = i-m-1$
② $\dim(|\partial\xi| \cap X_0) \leq i-m-2$

If $i \leq m$, can't intersect $X_0 = \{x\} \Rightarrow IC_i^{\underline{m}}(X) = C_i(X-x)$

If $i \geq m+2$, no condition $\Rightarrow IC_i^{\underline{m}}(X) = C_i(X)$

If $i = m+1$, ξ can go through x, but not $\partial\xi$

$$IC_\bullet(X): \cdots \to C_{m+3}(X) \to C_{m+2}(X) \to IC_{m+1}(X) \to C_m(X-x) \to C_{m-1}(X-x) \to \cdots$$

Result is clear for $i \neq m, m+1$.

$$\mathrm{Ker}\left[IC_{m+1}(X) \to C_m(X-x) \right] = \mathrm{Ker}\left[C_{m+1}(X) \to C_m(X) \right]$$

($IC_{m+1}(X) \cap C_{m+1}(X)$; since if $\partial\xi = 0$, $x \in |\partial\xi|$)

$\Rightarrow$ result holds at $m+1$

$$\partial : IC_{m+1}(X) \to \partial C_{m+1}(X) \cap \underbrace{IC_m(X)}_{C_m(X-x)}$$

$\Rightarrow IH_n$ is image of $H_n(X-x)$ in $H_n(X)$ //

⑥ **Cor.** X = pinched torus $= \{y^2 = x^3 - x^2\} \subseteq \mathbb{P}^2$ has

$$IH_i(X, \mathbb{Z}) = \begin{cases} \mathbb{Z} & i = 0, 2 \\ 0 & i = 1 \end{cases}$$

Pf: $H_1(X - x, \mathbb{Z}) \cong \mathbb{Z} \longrightarrow H_1(X, \mathbb{Z})$

($X - x \simeq S^1$)

$1 \longmapsto 0$

bc gen. is a bdry in X



//

~~Will need a perversity function at all~~ ← Will return to this later

---

- Computing with chains is hard; often computed via normalizations.

Normalization: "a type of partial compactification"

Def. A domain A is *normal* if it is integrally closed in its field of fractions.

i.e. $F = \mathrm{Frac}(A) \supseteq A$. A is normal if for every $a_1, \dots, a_m \in A$ & every $x \in F$ s.t. $x^m + a_1 x^{m-1} + \cdots + a_m = 0$ we have $x \in A$.

Def. The normalization of A is

$$\{x \in F \mid x \text{ satisfies } x^m + a_1 x^{m-1} + \cdots + a_m = 0 \text{ for some } m \text{ \& } a_i \in A\}.$$

⑦ Ex: $\{x^2 = y^3\} \subseteq \mathbb{C}^2$ (has singularity at (0,0))

$\Updownarrow$

$A = \mathbb{C}[x,y]/(x^2-y^3) \cong \mathbb{C}[t^3, t^2] \subseteq \mathbb{C}[t]$

$\mathrm{Frac}(A) \cong \mathbb{C}(t)$

t satisfies $t^3 - (x) = 0$ & $\overline{A} = \mathbb{C}[t]$

$A \hookrightarrow \overline{A} \iff \{x^2 = y^3\} \leftarrow \mathbb{A}^1$

(not equiv. to resolutions of singularities)

Normality is related to resolutions of singularities

Prop: If A is normal (noetherian), then A is "regular in codim 1"

ie. for every height 1 prime $\wp \subseteq A$, $A_\wp$ is regular.

(Geometrically, the singular locus of A is codim ≥ 2.)

Pf: Since $A_\wp$ is normal, it is sufficient to show:

Claim: If A is a normal, noetherian local ring of dimension 1, then its maximal ideal $m \subset A$ is principal.

Pf of Claim: $x \in m$ nonzero $\Rightarrow m = \sqrt{(x)} \Rightarrow m^n \subset (x)$ some n

Choose n smallest s.t. $m^n \subseteq (x)$, so $\exists y \in m^{n-1}$ s.t. $y \notin (x)$

$\Rightarrow t = \frac{y}{x} \in \mathrm{Frac}(A) \setminus A$ & $t \cdot m \subset A$ so $tm = A$ or $\subset m$.

If $tm \subset m \Rightarrow tm$ is f.g. A-module & is $A[t]$-module

$\Rightarrow t$ is integral $/A \Rightarrow$ contradiction.

So $t \cdot m = A$ & $\exists m \in m$ s.t. $tm = 1 \Rightarrow m = A \cdot t^{-1} = A \cdot m$

is principal //

So the normalization is a "partial resolution".

What does normality look like?

Def. X is topologically normal if $\forall x \in X$, $\exists$ open nghd $U_x \ni x$ s.t. $U_x - X_{n-2} = U_x - \mathrm{Sing}(X)$ is connected.

Ex:  is not topologically normal, but  is.

⑧ **Prop.** X normal $\Longrightarrow$ X topologically normal.

Pf: We use a key property of normal varieties:

Zariski's Main Thm: If $f: Z \to X$ is a proper, birat'l map; X is normal & Z is irreducible, then ① $f^{-1}(x)$ is connected $\forall x \in X$, ② $f_* \mathcal{O}_Z = \mathcal{O}_X$.

Hironaka's Thm: Any quasiprojective variety X admits a (proper) resolution of singularities $f: Z \to X$.

We argue by passing to Z: $x \in U$ (analytic) nghd of x. Then put $E = \pi^{-1}(U) \cap \pi^{-1}(\mathrm{Sing}(X))$. E is clsd in Z & Z is smooth, so $\pi^{-1}(U) \setminus E$ is still connected if $\pi^{-1}(U)$ is connected (in analytic top.) (bc $\pi^{-1}(U) = V_1 \sqcup V_2 \Rightarrow \pi(V_1) \cap \pi(V_2) = \emptyset$ because f is a resolution of singularities $\Rightarrow U = \pi(V_1) \sqcup \pi(V_2)$) //

$\| 12$

$U \setminus \pi(U)$

Thm (Goresky–MacPherson) $\pi: \tilde{X} \to X$: the normalization of X. Then $\pi_*: IH^{\bar{p}}_i(\tilde{X}; \mathbb{Z}) \to IH^{\bar{p}}_i(X; \mathbb{Z})$ is an isom.

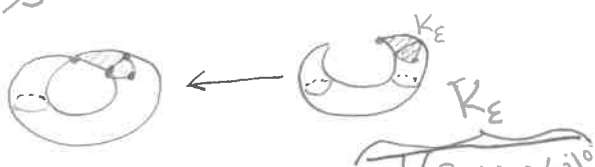
Pf: π induces a map on complexes $\pi_*: IC^{\bar{p}}_i(\tilde{X}, \mathbb{Z}) \to IC^{\bar{p}}_i(X;$ which takes a chain $\sigma: \Delta^i \to \tilde{X}$ to $\pi \circ \sigma$.

If $X_n \supseteq X_{n-1} \supseteq \cdots \supseteq X_0$ is strat of X, we can take preims. to get $\tilde{X}_n \supseteq \tilde{X}_{n-1} \supseteq \cdots \supseteq \tilde{X}_0$, which suffices to compute $IC^{\bar{p}}_i(\tilde{X};$ (Exercise: $\tilde{X}_n \supseteq \cdots \supseteq \tilde{X}_0$ is a Whitney stratification) ← See Back

We claim any chain $\sigma: \Delta^i \to X$ lifts uniquely to $\tilde{X}$

Since $\sigma(\text{Interior of } \Delta^i) \subseteq X^{\text{smooth}}$ & π is isom. over X^{smo} there is unique:

$$\begin{array}{ccc} (\Delta^i)^\circ & \xrightarrow{\tilde{\sigma}^\circ} & \tilde{X} \\ \cap & & \downarrow \\ \Delta^i & \xrightarrow{\sigma} & X \end{array}$$



For $a \in \Delta^i$, $B_\varepsilon(a) \cap (\Delta^i)^\circ$ is conn'd, so $\pi^{-1}(B_\varepsilon(a) \cap (\Delta^i)^\circ)$ is connected. $K_\varepsilon \subseteq \pi^{-1}(\sigma(\Delta^i))$: cpt (bc π is proper) $\supseteq K_\varepsilon \subseteq \pi^{-1}$ (from mu pair is)

$\bigcap_{\varepsilon > 0} K_\varepsilon$ is non-$\emptyset$ & conn'd (bc they are nested $\neq \emptyset$ compact sets) ⇝ so & intersection is a pt. & can lift uniquely &

↑ not true for arbitrary resolutions of singularities

8.5

Note on functoriality of IH:

Prop. Suppose $f: Y \longrightarrow X$ is a finite map. Then there are Whitney stratifications

$$Y = Y_n > Y_{n-1} > \cdots > Y_0 \quad \& \quad X = X_n > X_{n-1} > \cdots > X_0$$

s.t. $Y_i = f^{-1}(X_i)$.

Pf: Build stratifications together.

At each stage of induction, remove points, their preimages, & their images whenever they fail the 2 Whitney conditions. As this is a closed condition, we can then continue the induction downwards on dimension. //

This has applications beyond normal varieties!

e.g. $q: Y \longrightarrow Y/G$ for a finite group G acting on a normal variety Y.

Exercise: Show $IH^p_*(Y/G, \mathbb{Q}) = IH^p_*(Y, \mathbb{Q})^G$.

⑨ **Ex:** Back to Whitney umbrella:

$X = \mathrm{Spec}(A) = \{x^2 = y^2 z\} \subseteq \mathbb{C}^3$

$$A = \mathbb{C}[x,y,z]/(x^2 - y^2 z) \hookrightarrow \mathbb{C}[u,v]$$

$x \mapsto uv$

$y \mapsto u$

$z \mapsto v^2$

$A \cong \mathbb{C}[u, uv, v^2]$ ← (Can get v from $\lambda^2 - (v^2)$)

$\overline{A} = \mathbb{C}[u,v]$

$IH(X;\mathbb{Z}) = IH(\mathbb{C}^3;\mathbb{Z})$

For projective varieties, can glue normalizations for affine charts.

e.g. $X = \{x^2 w = y^2 z\} \subseteq \mathbb{P}^3$

On $w \neq 0$: $\nu_w : \mathbb{A}^2_{u,v} \to X$, $(u,v) \mapsto [uv : u : v^2 : 1]$

On $z \neq 0$: $\nu_z : \mathbb{A}^2_{a,b} \to X$, $(a,b) \mapsto [a : ab : 1 : b^2]$

glued where $v \neq 0$ / $b \neq 0$ with

$$\left[\tfrac{u}{v} : \tfrac{u}{v^2} : 1 : \tfrac{1}{v^2}\right] = [a : ab : 1 : b^2]$$

$a = \frac{u}{v}$, $b = \frac{1}{v}$

$\mathrm{Tot}(\mathcal{O}_{\mathbb{P}^1}(1))$ $\mathbb{P}^1$ gluing

+ $w = z = 0$ compactifies $\mathrm{Tot}(\mathcal{O}_{\mathbb{P}^1}(1))$

$\Rightarrow \overline{X} = \mathbb{P}(\mathcal{O}_{\mathbb{P}^1} \oplus \mathcal{O}_{\mathbb{P}^1}(1))$ "Hirzebruch surface"

$$IH_i(X,\mathbb{Q}) = H_i(\overline{X},\mathbb{Q}) \overset{\text{projective bundle formula}}{\underset{\text{[See back]}}{\cong}} H_i(\mathbb{P}^1,\mathbb{Q}) \oplus H_{i-2}(\mathbb{P}^1,\mathbb{Q})$$

$$= \begin{cases} \mathbb{Q}, & i=0 \\ 0, & i=1 \\ \mathbb{Q}\oplus\mathbb{Q}, & i=2 \\ 0, & i=3 \\ \mathbb{Q}, & i=4 \end{cases}$$

Projective Bundle formula: $\mathcal{E} \to X$: vector bundle.

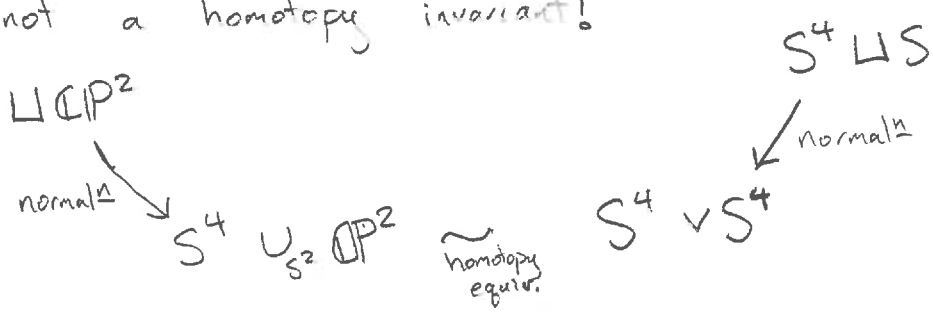
$$H^*(\mathbb{P}(\mathcal{E}); \mathbb{Z}) \cong \frac{H^*(X;\mathbb{Z})[\xi]}{\left(\xi^{r+1} + c_1(\mathcal{E})\xi^r + c_2(\mathcal{E})\xi^{r-1} + \cdots + c_{r+1}(\mathcal{E})\right)}$$

As $H^*(X;\mathbb{Z})$-algebras

Cor: $H^k(\mathbb{P}(\mathcal{E});\mathbb{Z}) \cong \bigoplus_{i=0}^{r} H^{k-2i}(X;\mathbb{Z})$ $\forall k$.

Rmk: IH is not a homotopy invariant!

e.g. $S^4 \sqcup \mathbb{CP}^2$ and $S^4 \sqcup S^4$



$$\Rightarrow IH_i^p(S^4 \vee S^4, \mathbb{Z}) \cong H_i(S^4, \mathbb{Z}) \oplus H_i(S^4, \mathbb{Z}) \quad \text{different!}$$
$$IH_i^p(S^4 \cup_{S^2} \mathbb{CP}^2, \mathbb{Z}) \cong H_i(S^4, \mathbb{Z}) \oplus H_i(\mathbb{CP}^2, \mathbb{Z})$$

Note: In both formulas above, answer is independent of p! This reflects smoothness of normalization, as we prove below:

Thm. X: oriented, normal, cpt n-dim'l pseudomfd / X: projective, normal (real dim n) complex **variety**.

Then: $IH_i^{\underline{t}}(X, \mathbb{Z}) \cong H_i(X, \mathbb{Z})$
$IH_i^{\underline{o}}(X, \mathbb{Z}) \cong H^{n-i}(X, \mathbb{Z})$

[$\underline{o}$ = 0-perversity fxn
$\underline{t}$ = top perversity fxn
$\underline{t}(i) = i-2$, $i \geq 2$.]

[**Rmk.** Other perversities "interpolate" between H_i & H^{n-i}.]

Pf Ideas: **LemA:** $IH_0^p(Y) = H_0(Y^{sm})$

[Compute w/ chains: $\dim(|\xi| \cap Y_{n-2}) \leq -2$ for $\xi \in IC_0^p$
& $\dim(|\xi| \cap Y_{n-2}) \leq -1$ for $\xi \in IC_1^p$]

$\Rightarrow IH_0^p$ same as computing H_0 for $Y^{sm} = Y - Y_{n-2}$.

MOVE TO LATER

LemB L: compact of dim. $k-1$, with Whitney strat^n $L = L_{k-1} \supset L_{k-2} \supset \cdots \supset L_0$

$$IH_i^p(\text{cone}(L)) = \begin{cases} IH_i^p(L), & i < k-1-p_k \\ 0, & i \geq k-1-p_k \end{cases}$$

Pf: $\text{cone}(L)$ has stratif^n $\text{cone}(L)_i = \begin{cases} \text{cone}(L_{i-1}), & i \geq 1 \\ \{v\} \ (\text{cone point}), & i = 0 \end{cases}$

$\xi \in C_i(\text{cone}L)$ is allowable at v iff:
$0 \leq i - k + p_k \iff i \geq k - p_k$

If $i \leq k - p_k - 1$, $\xi \in IC_i^p(\text{cone}L)$ can't touch v &

$$IC_i^p(\text{cone}(L)) \cong IC_i^p(L \times (0,1)) \cong IC_i^p(L) \Rightarrow \text{for } i < k-1-p_k,\ IH_i^p(\text{cone}L) = IH_i^p(L)$$

(11) If $i \geq k - p_k - 1$, form $c\xi = \mathrm{cone}(\xi)$ for $\xi \in IC^p_i(\mathrm{cone}\, L)$

By hypothesis, $\dim(|c\xi| \cap \{v\}) = 0 \leq (i+1) - k + p_k$

& $\dim(|c\xi| \cap \mathrm{cone}(L_{k-1-j})) \leq \dim(|\xi| \cap L_{k-1-j}) + 1$

$\leq i - j + p_j + 1$

So $c\xi \in IC^p_{i+1}(\mathrm{cone}\, L)$.

Easy to check: $\partial(c\xi) = \xi \Rightarrow IH^p_i(X) = 0$ for $i \geq k - p_k - 1$ //

Lem: $IH^p_*(\mathbb{R}^m \times Y) \cong IH^p_*(Y)$

[Pass to normalizations & split as product. Use $H_*(\mathbb{R}) \cong \mathbb{R}$ in deg 0

Mayer-Vietoris for IH: For $U, V \subseteq X$ open, we have a s.e.s of complexes:

$$0 \to IC^p_*(U \cap V) \xrightarrow{\xi \mapsto (\xi, \xi)} IC^p_*(U) \oplus IC^p_*(V) \xrightarrow{(\alpha,\beta) \mapsto \alpha - \beta} IC^p_*(U \cup V) \to 0$$

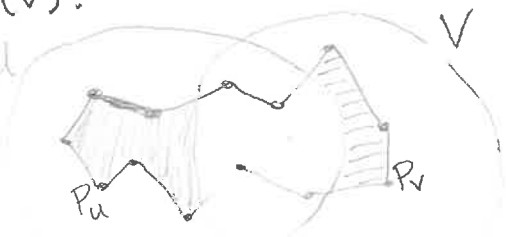
[so we have a l.e.s.

$$\cdots \to IH^p_i(U \cap V) \to IH^p_i(U) \oplus IH^p_i(V) \to IH^p_i(U \cup V) \xrightarrow{\partial} IH^p_{i-1}(U \cap V) \cdots$$

Pf: Most is formal from usual MV sequence.

Hardest Part: If $\xi \in IC^p_i(U \cup V)$, then $\xi = \xi_U + \xi_V$ for $\xi_U \in IC^p_i(U)$ & $\xi_V \in IC^p_i(V)$.

Pf Sketch:



Shrink simplices enough s.t. no simplex intersects both P_U & P_V. AND $|\xi| \cap X_i$ are sub-complexes

define $P = |\xi|$

$f: P \to \mathbb{R}$ cts. s.t.

$f(v) = \begin{cases} 0 & \text{if } v \in \{\text{clsd simplices intersecting } P_U\} \\ 1 & \text{if } v \in \{\text{clsd simplices intersecting } P_V\} \\ \frac{1}{2} & \text{else} \end{cases}$

for vertices v & extending affinely to P.

Pick $c \in (0,1)$ avoiding vertices, & ξ_U = part of ξ w/ $f < c$ & ξ_V = part of ξ w/ $f > c$.

Compatibility is clear for $|\xi_U|$ & $|\xi_V|$.

$|\partial \xi_U| \subseteq |\partial \xi| \cup f^{-1}(c)$

"Thin Wall Lemma": For every subpolyhedra $Y \subseteq P$, $\dim(f^{-1}(c) \cap Y) \leq \dim(Y) - 1$

//

(12) # Pf of $IH^{\bar{p}}$ & $IH^{\bar{t}}$ for normal X:

Claim: If F & G : {open sets in X} $\longrightarrow$ {graded abelian gps} both satisfy:

① $F(U) \longrightarrow F(V)$ for $U \hookrightarrow V$

② F & G satisfy M.V. exact sequences

③ F & G commute w/ direct limits over increasing unions.

Then if $\phi : F \longrightarrow G$ is a natural transformation (which is an isomorphism when $V \cong \mathbb{R}^n \subset X^{sm}$ or $V \cong \mathbb{R}^{n-k} \times \text{cone}(L)$, then ϕ_X is an isomorphism.

Pf: Use MV sequences: • if ϕ is an isom. on U, V, $U \cap V$, then $\phi_{U \cup V}$ is isomorphism.

• Induct on Whitney stratification, covering X_{n-i} at each step.

Thm Part 1: $IH_i^{\bar{t}}(X, \mathbb{Z}) \cong H_i(X, \mathbb{Z})$

Pf: need to check for cone(L):

i	$IH_i^{\bar{t}}(\text{cone}\,L)$	$H_i(\text{cone}\,L)$
≥ 1	$\to 0$	0 [cone L is contractible]
0	$IH_0^{\bar{t}}(L) = H_0(L^{sm})$	$\mathbb{Z}$

$\dim(|\xi| \cap X^{sing}) \leq i-2$ $\forall$ coh degs. k. $\Rightarrow IC_*^{\bar{t}} \leq C_*^{PL}$ $\Rightarrow IH_*^{\bar{t}} \to H_*$

$k-1-(k-2) = 1$, so follows from Lems B, then A

Agree when L^{reg} is connected $\Leftarrow$ top. normality. //

Thm Part 2: $IH_i^{\bar{p}}(X, \mathbb{Z}) \cong H^{n-i}(X; \mathbb{Z})$

map comes from $\sigma \longmapsto D(\sigma)$ = dual block of barycentric subdivision.

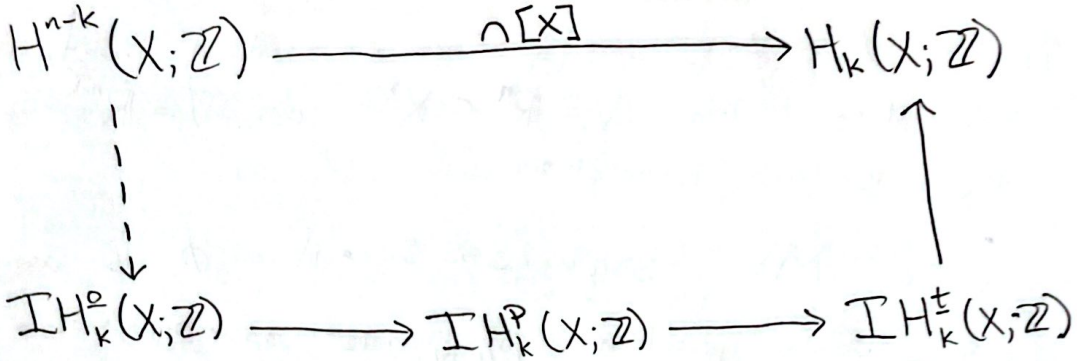
↑ This map requires starting with a triangulation! Better: use the cap product. See Buck.

$V = \mathbb{R}^{n-k} \times \text{cone}(L)$:

i	$IH_i^{\bar{p}}(V)$	$H_c^{n-i}(V)$
$\leq k-2$	$IH_i^{\bar{p}}(L)$ ← induction →	$H^{k-1-i}(L)$
$k-1$	$\to 0$ ← when L is connected, i.e. normality →	$\tilde{H}^0(L)$
$\geq k$	$\to 0$	0

$k-1-0 = k-1$, so Lem B computes

Prop. [Cap product factors through IH]: $\forall$ perversity fcn $\bar{p}$, there is a factorization



- Can be proved using a triangulation of X + explicit formula for cap product.
- We will defer proof as it follows more cleanly from sheaf-theoretic constructions.

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(12.5) Borel-Moore Homology & $I^{BM}H$:

(-Introduce/motivate BM homology
-MV for BM
-Compute for cones.

X: smooth, but non-compact, oriented mfds.

→ duality & many applications of cohomology require compact supports

eg. $H^i_c(X,\mathbb{Z}) \cong H_{n-i}(X,\mathbb{Z})$

$H^i(X,\mathbb{Z}) \cong H^{BM}_{n-i}(X,\mathbb{Z})$

"Borel-Moore" → generalizes well to IH

Borel-Moore Homology: Replaces singular chains $\sum_{i=1}^{k} a_k \cdot \sigma_k$ with infinite sums $\sum a_\sigma \cdot \sigma$ s.t. $\forall$ compact $K \subseteq X$, only finitely many σ intersect K.

Ex: for $\mathbb{R}$, we have $\sum_{i\in\mathbb{Z}} [i,i+1] \in C_1((\mathbb{R}))$

Can use a triangulation: $\mathbb{R} = \bigcup_{i\in\mathbb{Z}} [i,i+1]$.

$$0 \longrightarrow \prod_{n\in\mathbb{Z}} \mathbb{Z} \longrightarrow \prod_{n\in\mathbb{Z}} \mathbb{Z} \longrightarrow 0$$

$$(a_n) \longmapsto (a_{n+1}-a_n)$$

$H_1 = \mathbb{Z}\cdot(1)_{n\in\mathbb{Z}}$ $\quad H_0 = 0$

Fact: If $\overline{X}$ is cpt'n of X & $D = \overline{X} - X$,

$H^i_c(X,\mathbb{Z}) \cong H^i(\overline{X}, D; \mathbb{Z})$

If M is compact, it is clear (from chains) that $H_i = H^{BM}_i$.

Can do the same modification to get $I^{BM}H^p_i$. All results we discussed for IH hold for $I^{BM}H$ (compute by normalizations, MV sequences, etc.).

Goal: $I^{BM}H^p_i(cL) = ?$ for $\dim(L) = k-1$; L: compact (w/o bdry) pseudomfd.

(cL = cone(L))

SubLem: $I^{BM}H^p_i(X;\mathbb{Z}) \cong I^{BM}H^p_{i+1}(X\times\mathbb{R};\mathbb{Z})$

Pf: Still splits as a product, but $I^{BM}H^p_i(\mathbb{R},\mathbb{Z}) = \begin{cases} 0 & , i=0 \\ \mathbb{Z} & , i=1 \end{cases}$

Prop $I^{BM}H^p_i(cL,\mathbb{Z}) = \begin{cases} IH^p_{i-1}(L,\mathbb{Z}) & , i > k-1-p_k \\ 0 & , i \le k-1-p_k \end{cases}$ ↙ inequalities reversed from standard case!

Pf: Same condition at cone point: $\dim(|\xi| \cap \{0\}) \le i-k+p_k$

so no intersection when $i \le k-1-p_k$ [when $i-k+p_k \le -1$]

For $\xi \in IC^p_i(L\times(0,\infty))$ (missing 0) form $R = \sum_{m\ge 0} [m,m+1]$ on $[0,\infty)$ & $\xi\times R$.

& consider $\eta = F_*(\xi\times R)$: composing $\xi\times R$ with $F: L\times(0,\infty)\times[0,\infty) \xrightarrow{\pm} L\times(0,\infty)$

$\partial(\xi\times R) = \partial\xi \times R + (-1)^i \xi \times \partial R \overset{\partial\xi=0}{=} (-1)^{i+1} \xi \times \{0\} \Rightarrow \partial\eta = \xi$

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So this "coning off to infinity" procedure shows

$$I^{BM}H_i^P(cL) = 0 \quad \text{for} \quad i \leq k-1-p_k.$$

For higher degrees $\xi \in IC_i^P(L) \Rightarrow c_\infty(\xi) = \overline{\xi \times \mathbb{R}} \approx \xi \times [0,\infty)$

$$c_\infty(\xi) \in I^{BM}C_{i+1}^P(L)$$

$$c_\infty : IC_*^P(L) \hookrightarrow IC_{*+1}^P(cL)$$

for $*$ large enough to go through cone pt.

$$\partial(c_\infty \sigma) = -c_\infty(\partial\sigma)$$

//

(13) # Poincaré Duality for IH:

Recall: $\overline{t}(i) = \begin{cases} 0, & i \le 2 \\ i-2, & i > 2 \end{cases}$ is the maximal/top perversity and

$\underline{0}(i) = 0 \ \forall i$ is the zero perversity.

▷ For p: perversity fcn, define $q = \overline{t} - p$ to be the "complementary perversity".

Thm: X: projective complex variety; p & q = complementary. Then there is a non-degenerate bilinear pairing

$$IH^p_i(X;\mathbb{Q}) \times IH^q_{n-i}(X,\mathbb{Q}) \xrightarrow{\cap} \mathbb{Q}.$$

Notes: ▷ similar versions when X is not projective adding suitable "cpt. supports" conditions.

▷ when X is smooth this is constructed by taking $\sigma_i : \Delta^i \to X$ & $\sigma_{n-i} : \Delta^{n-i} \to X$, nudging them until they are transverse, & sending to $\#(\sigma_i \cap \sigma_{n-i})$. Proof of non-degeneracy comes from local computations + M.V. sequence

Idea for proof is similar: take chains s.t. $|\sigma_i| \cap |\sigma_{n-i}| \subseteq X^{sm}$ & is finitely many points. Using M.V., reduces to showing the corresponding statement for cones $\text{cone}(L)$.

Recall: for $\dim(L) = m$,

$$IH^p_i(\text{cone}\,L, \mathbb{Z}) = \begin{cases} IH^p_i(L;\mathbb{Z}), & i < m - p_{m+1} \\ 0, & \text{else} \end{cases}$$

$$I^{BM}H^p_i(\text{cone}\,L, \mathbb{Z}) = \begin{cases} 0, & i \le m - p_{m+1} \\ IH^p_{i-1}(L;\mathbb{Z}), & \text{else} \end{cases}$$

(copy before class)

If L satisfies P.D. (induction on dim), then $\forall i < m - p_{m+1}$

$$IH^p_i(\text{cone}\,L, \mathbb{Q}) \cong IH^p_i(L, \mathbb{Q}) \xrightarrow{\cong} IH^q_{n-i}(L, \mathbb{Q})^* \cong I^{BM}H^q_{m+1-i}(\text{cone}\,L, \mathbb{Q})^*$$

& if $i \ge m - p_{m+1}$, since $m - 1 = p_{m+1} - q_{m+1}$,

$$IH^p_i(\text{cone}\,L, \mathbb{Q}) = 0 = I^{BM}H^q_{m+1-i}(\text{cone}\,L, \mathbb{Q})$$

Later, we'll return & reprove this with sheaves / Verdier duality.

(13.5) Rmk: Here, the complentary perversities appear in local calculations (for cones). Another way to see complementary perversity fcns must appear is through the heuristics of intersection #'s:

an allowable i-chain σ_i & $n-i$ chain σ_{n-i} must not meet on X_{n-k}. Their expected intersection dim. is

$$(i-k+p_k)+(n-i-k+q_k)-(n-k)=p_k+q_k-k.$$

To make this <0, need $p_k+q_k \leq k-1$

For homologies to also miss each other: $p_k+q_k \leq k-2$

Making this equal allows "as many cycles in dual as possible" $\longrightarrow$ makes pairing non-degenerate.

(14) Note there a 2 "middle" perversities:

$\underline{m} = [0,0,0,1,1,2,2,3,3,\dots]$ "upper middle"

$\underline{n} = [0,0,0,0,1,1,2,2,3,3,\dots]$ "lower middle"

$m + n = \underline{t}$ are complementary.

For general pseudomanifolds (locally conical mfds), $IH^{\underline{m}} \neq IH^{\underline{n}}$

e.g. real algebraic varieties don't necessarily have $IH^{\underline{m}} = IH$

Prop. If all strata in a Whitney stratifn of X are even dimensional, then $IH^{\underline{m}}_*(X) = IH^{\underline{n}}_*(X)$.

ex: X: complex algebraic variety.

Pf: Since strata are even-dim'l, they only see $p(2i)$ for $0 \le i \le \dim_{\mathbb{C}}(X) = \frac{1}{2}\dim_{\mathbb{R}}(X)$. //

In this case, $IH^{\underline{m}}_*(X) = IH_*(X)$ is self-dual under Poincaré Duality.

e.g. If X: cx alg. variety has smooth normalization, then $IH^{\underline{p}}_*(X) = IH_*(X)$ $\forall$ perversity fcns p & so $IH^{\underline{p}}_*(X)$ satisfy Poincaré Duality.

Introduces new invariants associated to X

e.g. If X has even complex dimension $2n$, get a non-deg. pairing $IH^{\underline{m}}_{2n}(X,\mathbb{Q}) \otimes IH^{\underline{m}}_{2n}(X,\mathbb{Q}) \longrightarrow \mathbb{Q}$.

$\sigma(X) :=$ #(positive eigvals) − #(negative eigvals)

"signature of X" $\longrightarrow$ topological (bordism) invariant of X.

(15) # Crash Course on Sheaf Theory:

New Goal: Sheafify everything we talked about.

$IH_i \rightsquigarrow IC$ sheaves

We want this for a few reasons:

- ▷ Can start to deal with families.
- ▷ Can understand relationships between IC sheaves on subvarieties of ambient spaces / sources & targets of maps
- ▷ Categorifies constructions in geometric rep. theory, number theory / Langlands program, etc.

Def. A presheaf is a contravariant functor

$$\mathcal{F}: \{\text{open sets in } X\} \longrightarrow \{A\text{-modules}\}$$

(e.g. abelian gps.; $\mathcal{O}_X$-modules; vector spaces; ...)

$$\mathcal{F}(U) =: H^0(U, \mathcal{F}) =: \Gamma(U, \mathcal{F})$$

$\mathcal{F}$ is a sheaf if it also satisfies the gluing property: • If $s_i \in \Gamma(V_i, \mathcal{F})$ s.t. $s_i|_{V_i \cap V_j} = s_j|_{V_i \cap V_j}$ $\forall i,j$

Then $\exists!\ s \in \Gamma(\bigcup_i V_i, \mathcal{F})$ s.t. $s|_{V_i} = s_i$ $\forall i$

[Note: might be infinitely many V_i !]

e.g. ① $M \in A\text{-mod}$.

$\underline{M}_X$ = constant sheaf on X

$\underline{M}_X(U) := \{\text{cts. } f: U \to M\} \cong M^{\#\pi_0(U)}$

② "Structure sheaf" of X, $\mathcal{O}_X = \{\text{sheaf of fcns on } X\}$

If X is a scheme & $U = \mathrm{Spec}(R) \subset X$ is an affine patch, $\mathcal{O}_X(U) = R$.

Def. The stalk of $\mathcal{F}$ @ $x \in X$ is $\mathcal{F}_x = \varinjlim_{x \in U} \mathcal{F}(U)$

(16) There are 2 flavors of sheaves that will be interested in: coherent & constructible

Coherent: Any algebraic variety, X, comes with a structure sheaf $\mathcal{O}_X$ of fcns on X.

$$\mathcal{O}_X(U) = \{\text{algebraic fcns on } X\}$$

A coherent sheaf is a sheaf $\mathcal{F}$ on X s.t.

▷ $\mathcal{F}(U)$ is a fin. presented $\mathcal{O}_X(U)$-module for all small enough U

If $U = \mathrm{Spec}(A) \subseteq X$ is an affine (Zariski) open set, $\mathcal{O}_X(U) = A$ & $\mathcal{F}|_U = \tilde{M}$ for a f.g. A-module M (i.e. $\mathcal{F}(U_{-(a)}) = M_{(a)}$.)

NOTE: "Coherent" usually means Zariski topology!

Constructible:

- Def. An A-local system on X is a locally constant sheaf $\mathcal{L}$ of A-mods. i.e. $\exists$ open covering $X = \bigcup_i U_i$ s.t. $\mathcal{L}|_{U_i} \cong \underline{M_i}$ for an A-mod M_i.

▷ If X is connected $M_i = M$ $\forall i$ is constant.

▷ If $M = M_i = A^r$ $\forall i$, then $\mathcal{L}$ is of "rank r".

Exs: ① (from topology) $\mathcal{O}r_X^{naive}(U) = H_n(X, X-U; A)$

$\mathcal{O}r_X = (\mathcal{O}r_X^{naive})^{sh}$

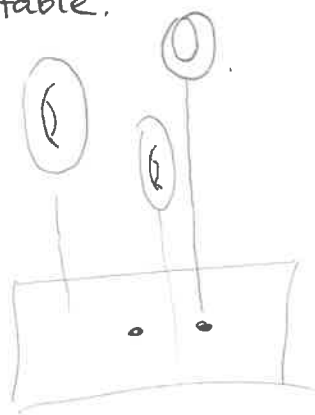
$\mathcal{O}r_X$ is constant $\iff$ X is orientable.

② (family) $\mathcal{X} = \{y^2 = x(x-1)(x-2)\}$

$f \downarrow$

$\mathbb{A}^1_\lambda \setminus \{0,1\}$

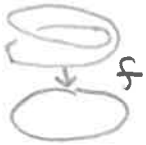
$\mathcal{F}(U) = H_1(f^{-1}(U), \mathbb{Z})$ } stalks are $\mathbb{Z}^2$



(17) How to tell what a locally constant sheaf is?
(Or even whether it is nontrivial??)

A: Walk along paths in X & see if you end up in the same place!

Ex: Consider double cover of S^1



$\mathcal{F}(U) := f_* \underline{A}_{S^1}(U) := \underline{A}_{S^1}(f^{-1}(U))$
is locally constant. $\mathcal{F}_x \cong A^2$ (rank 2 local system)

Can trivialize $S^1 = U_1 \cup U_2$

$\mathcal{F}|_{U_i} \cong \underline{A^2}_{U_i}$

$(a_1, a_2) \in \mathcal{F}(U_1) \longrightarrow (a_1, a_2) \in \mathcal{F}(U_2) \longrightarrow (a_2, a_1)$

$\longrightarrow \pi_1(S_1, x_0) \longrightarrow \mathrm{Aut}(A^2)$

$1 \longmapsto \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ order 2 image

Prop. X: connected top. space. Then TFAE:
(a) A-local systems on X w/ fibers M
(b) representations $\rho: \pi_1(X, x_0) \to \mathrm{Aut}_A(M)$.

Pf: ① $\mathcal{L}$: local system $\Rightarrow$ $\rho_{\mathcal{L}}: \pi_1(X, x_0) \longrightarrow \mathrm{Aut}_A(M)$

Bc: $p: [0,1] \longrightarrow X$ gives a trivial local system $p^{-1}(\mathcal{L})$ on $[0,1]$ & $p^{-1}\mathcal{L}_0 \cong \mathcal{L}_{p(0)}$, $p^{-1}\mathcal{L}_1 \cong \mathcal{L}_{p(1)}$.

② $\rho: \pi_1(X, x_0) \to \mathrm{Aut}_A(M)$ $\Rightarrow$ $\mathcal{L}_\rho$: local system

Bc: $\mathcal{L}$ = sheaf of sections of $(\tilde{X} \times M)/\pi_1(X, x_0)$
for $\tilde{X} \to X$: universal cover.

$\rho_{\mathcal{L}_\rho} = \rho$: Sheaf of sections of $(\tilde{X} \times M)/\pi_1(X, x_0)$ has monodromy:

$\gamma \in \pi_1(X, x_0)$ acts by $\cdot$ $\gamma(s(\tilde{x}_0)) = s(\tilde{\gamma}(1)) = s(g_\gamma \tilde{x}_0) = \rho(\gamma)\, s(\tilde{x}_0)$
$s: p^{-1}(U) \to M$ loc. const. s.t. $s(g \cdot \tilde{x}) = \rho(g) \cdot s(\tilde{x})$

$\mathcal{L}_{\rho_{\mathcal{L}}} = \mathcal{L}$: $\tau: M_{\tilde{x}} \xrightarrow{\sim} p^*\mathcal{L}$ $\circlearrowleft$ Deck gp. π_1, $\tau_{g\gamma\tilde{x}} = T_\alpha \circ T_\gamma = \tau_{\tilde{x}} \circ \rho(\gamma)$

(Harder example) Recall the elliptic curve example.

Consider $\mathcal{L}(u) = H_1(f^{-1}(u))$. ← fibers $\mathbb{Z}^2$ $\pi_1(\mathbb{A}^1_{\mathbb{C}} - \{0,1\}) \cong F_2$.

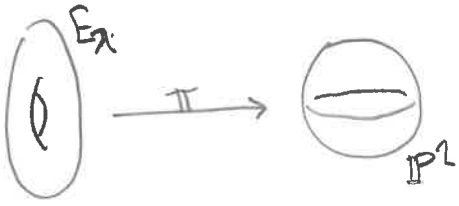
Q: What is $\rho: F_2 \longrightarrow \mathrm{Aut}(\mathbb{Z}^2)$ corresponding to $\mathcal{L}$?

A: $\rho(e_1) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$ "Dehn twist"



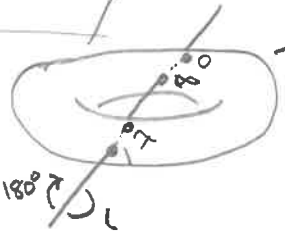
(more in Geordie's notes...)

Pf:





Topologically: π:



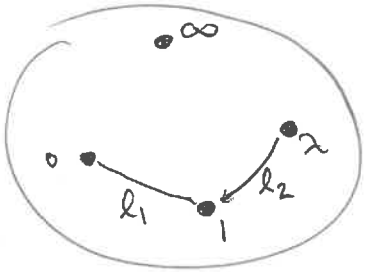
$\longrightarrow T/\iota \cong$

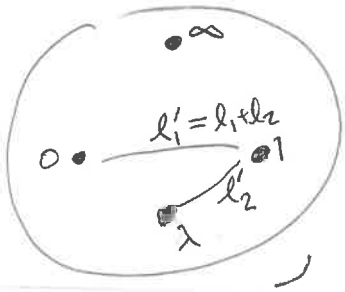


$\cong S^2$









In larger family $X = \{y^2 = (x+a)(x-b)(x-c)\}$

Pf. of Ehresmann's Thm: $f: X \to Y$: proper, submersion of smooth mfd

Ehresmann is local on target: suffices to prove for $Y = \mathbb{R}^n$

Submersions have local coordinates @ $x \in X$ given by

$\mathbb{R}^{n+m} \xrightarrow{\text{proj}} \mathbb{R}^n$.

Choose partition of unity for open cover of $X \Rightarrow$ lift $\partial_i \in \mathfrak{X}(\mathbb{R}^n)$

to $v_1, \dots, v_n \in \mathfrak{X}(X)$ $\left[v_i = \sum_\alpha \rho_\alpha \frac{\partial}{\partial x_i^{(\alpha)}}\right]$ s.t. $df_p(v_i) = \partial_i \ \forall p$

φ^i = flow along v_i ; $\Phi : \mathbb{R}^n \times f^{-1}(q) \to X$ by

$$\Phi(s_1, \dots, s_n, p) = \varphi^n_{s_n} \circ \cdots \circ \varphi^1_{s_1}(p)$$ is an isom.

[Uses properness to show flow is well-defined/doesn't blow up & easy to check $f(\Phi(s,p)) = s$.]

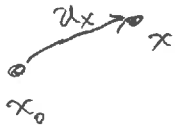
Lem: $H^c(X, \mathcal{L}) \cong M^{\pi_1(X, x_0)}$ when X is conn'd

(M = fiber of $\mathcal{L}$ @ x_0)

$$H_0(X, \mathcal{L}) \cong M_{\pi_1(X,x_0)} = M / \langle \gamma_* m - m \rangle$$

Pf: $$H_0(X, \mathcal{L}) = \left(\bigoplus_{x \in X} \mathcal{L}_x\right) \Big/ \langle (\alpha_* m)_{\alpha(1)} - m_{\alpha(0)} \mid \alpha = \text{path (1-simplex)} \ \& \ m \in \mathcal{L}_{\alpha(0)} \rangle$$

$\phi(m_x) = u_{x,*}^{-1}(m_x)$] use path to transport to x_0



$M_{\pi_1(X,x_0)}$

▷ ϕ is well-defined bc. $\gamma = u_x' \circ u_x^{-1}$ is loop

so $\gamma_* \circ u_{x,*}^{-1}(m_x) - u_{x,*}^{-1}(m_x) = 0$.

$\gamma_* \circ u_{x,*}^{-1}(m_x) = u_{x',*}^{-1}(m_x)$

▷ Relations exactly match after ϕ. //

Claim: $f: X \to Y$: smooth + proper $\Rightarrow$ $f_* \mathcal{L}$ is a local system if $\mathcal{L}$ is.

2 Main Ingredients:

① **Proper base change:** Suppose we have a Cartesian diagram:

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & \lrcorner & \downarrow{\scriptstyle f} \\ S' & \xrightarrow{g} & S \end{array}$$

If f is proper & $\mathcal{F}$ is constructible, then $g^* f_* \mathcal{F} \cong f'_* g^* \mathcal{F}$.

properness used for cptness of fibers & "tube lemma" [$\lim_{U \supseteq Y} \mathcal{F}(f^{-1}U) \cong \Gamma(X_y, \mathcal{F}|_{X_y})$]

Cor:

$$\begin{array}{ccc} X_s & \to & X \\ \downarrow & & \downarrow f \\ \mathrm{Spec}(k) & \xrightarrow{s} & S \end{array}$$

Then $(\mathcal{F}|_{X_s})(X_s) = f_* \mathcal{F}|_s$ for $s \in S$.

i.e. fiber of $f_* \mathcal{F}$ is $\Gamma(\mathcal{F}|_{\text{fiber}})$.

Note: proper is necessary! $f: \mathbb{A}^2 \setminus \{(0,0)\} \xrightarrow{\pi_2} \mathbb{A}^1$

$f_* \mathcal{O}_{\mathbb{A}^2 - \{0\}} \cong \mathcal{O}_{\mathbb{A}^1}[x]$ & $\Gamma(\mathcal{O}_{\mathbb{A}^1 - \{0\}}, \mathbb{A}^1 - \{0\}) = k[x, x^{-1}]$

$\Rightarrow f_* \mathcal{O}|_0 = k[x]$ ← not equal!

MORE ON BACK!

② **Ehresmann's Thm:** A proper submersion of smooth mfds is a locally trivial fibration, **ie.** $\forall y \in Y$, $\exists$ (analytic) nghd $\mathcal{U} \ni y$ s.t. $f^{-1}(\mathcal{U}) \cong \mathcal{U} \times f^{-1}(y)$.

~~By ②: $\mathcal{L}|_{X_y}$ is locally trivial & ...~~

▷ Can argue claim only with ② actually! ① is just needed to say $f_* \mathcal{L}$ measures $H^i(X_y, \mathcal{L}|_{X_y})$.

Prop: $f: X \to Y$: proper map & $\mathcal{L}$: local system (or constructible sheaf) on X.
Then $f_* \mathcal{L}$ is constructible on Y.

Pf: uses a powerful result on Whitney stratif^{n}'s:

★ **Thom's first isotopy lemma:** If $X = \bigsqcup_\lambda X_\lambda$ is Whitney strat^{n} s.t. $X_\lambda \to Y$ is smooth, then $f: X \to Y$ is a stratified-locally-trivial fibration. } have non sing X

★ **Existence:** Build a Whitney strat^{n} of X & Y s.t.:
- ▷ $f^{-1}(Y_\lambda)$ is a union of strata of X
- ▷ $f_\mu: X_\mu \to Y_\lambda$ is a submersion/smooth when it is defined.
- ▷ ($\bigsqcup_\lambda X_\lambda$ has $\mathcal{L}|_{X_\lambda}$ locally constant.)

★ Then $f_* \mathcal{L}|_{Y_\lambda}$ is a local system by isotopy lemma $\Rightarrow f_* \mathcal{L}$ is constructible //

(Co)Homology with coefficients in a local system:

for $\sigma: \Delta^i \longrightarrow X$ & $\mathcal{L}$: local system $\Longrightarrow \sigma^*\mathcal{L}$: loc. sys. on Δ^i

Note Δ^i = contractible, so $\sigma^*\mathcal{L}$ = constant shf.

$\mathcal{L}_\sigma := \sigma^*\mathcal{L}(\Delta^i)$ = fiber of $\mathcal{L}|_{\sigma(\Delta^i)}$; $\forall p \in \Delta^i$, get rest^n map

$P_p^\sigma : \mathcal{L}_\sigma \xrightarrow{\sim} \mathcal{L}_{\sigma(p)} = (\sigma^*\mathcal{L})_p$

pf that σ: $P_\tau^\sigma = (P_p^\tau)^{-1} \circ P_p^\sigma : \mathcal{L}_\sigma \longrightarrow \mathcal{L}_\tau$

$$C_i(X,\mathcal{L}) = \left\{ \sum_\sigma l_\sigma \cdot \sigma \;\middle|\; \begin{array}{l} \Sigma \text{ is finite (modify for BM)} \\ l_\sigma \in \mathcal{L}_\sigma \end{array} \right\}$$

$$\partial_i\Big(\sum_\sigma l_\sigma \sigma\Big) = \sum_\sigma \sum_{\substack{\tau: \text{face} \\ \text{of } \sigma}} \pm \underbrace{P_\tau^\sigma(l_\sigma)}_{\text{only difference from normal homology.}} \cdot \tau$$

$$H_i(X,\mathcal{L}) = \ker(\partial_i) \Big/ \mathrm{Im}(\partial_{i+1})$$

Ex: $H_i(\mathbb{C}^\times, \mathcal{L}_m) = ?$ [$\mathcal{L}_m$ = local system for monodromy $z \mapsto z^m$]

e.g. $\mathbb{Z} \longrightarrow \mathrm{Aut}(A) = A^\times$, $n \longmapsto \lambda^n$

$\mathbb{C}^\times \underset{\text{homotopy}}{\sim} S^1$, so suffices to do S^1: $S^1 =$  has triangulation $T = \sigma: \Delta^1 \to S^1$

$$C_1^T(\mathbb{C}^\times, \mathcal{L}_m) = C_0^T(\mathbb{C}^\times, \mathcal{L}_m) = A.$$

the isomorphism matters!

$$0 \longrightarrow A \xrightarrow{\lambda - Id_A} A \longrightarrow 0$$

$$H_0(\mathbb{C}^\times, \mathcal{L}_m) = \ker(\zeta_m - 1) = \begin{cases} 0 & \text{if } m \neq 1 \\ k & \text{if } m = 1 \end{cases}$$

$$H_1(\mathbb{C}^\times, \mathcal{L}_m) = k/(\zeta_m - 1) = \begin{cases} 0 & \text{if } m \neq 1 \\ k & \text{if } m = 1 \end{cases}.$$

▷ Can define similarly for IH_i

▷ Have all similar formulas as H_i: MV seqs, [Poincaré duality], if fibers are v.s.: $IH_i(X,\mathcal{L}) \cong IH^i(X,\mathcal{L})^*$

▷ $C_*(X,\mathcal{L}) = C_*(X_H) \otimes_{G/H} M$ & $C^*(X,\mathcal{L}) = \mathrm{Hom}^*_{G/H}(C_*(X_H), M)$.

$H \subset \ker(\rho_x)$ ↑ fiber

▷ $H^0(X;\mathcal{L}) \cong M^{\pi_1(X)}$.

(19) local systems are clsd under $\oplus$, $\otimes$, $\underline{Hom}$, $(-)^\vee = \underline{Hom}(-, \underline{\Delta}_X)$

In fact, they are even better:

Prop: $Loc(X) \subset Sh(X)$ is a full abelian subcat.

Pf: $\mathcal{L} \xrightarrow{\phi} \mathcal{L}'$ a map of local systems.

$Ker(\phi)(U) := Ker(\underbrace{\mathcal{L}(U) \xrightarrow{\phi(U)} \mathcal{L}'(U)}_{\text{locally constant.}})$ ← no sheafification

$Coker(\phi)(U) := Coker[\mathcal{L}(U) \to \mathcal{L}'(U)]^{Sh}$

↖ Sheafification

Note: f_* is simple as coherent case, but f^* is not!

$f^*\mathcal{F} := f^{-1}\mathcal{F}$

$f^{-1}\mathcal{F}|_V = \underline{M}_V$ for $V = f^{-1}(U)$ s.t. $\mathcal{F}|_U = \underline{M}$

Note: **NOT** true for vector bdles for example:

On $\mathbb{P}^1$, have $0 \to \mathcal{O}(-1) \to \mathcal{O} \to \mathcal{O}_{pt} \to 0$.

Prop. $Loc(X)$ is stable under f^* & f_* for finite covering maps.

WARNING: **Not** stable under f_* for general proper maps!

(fibers can jump for example!)

Constructible Sheaves: These will capture $Rf_* \underline{k}_X$ $\forall f: X \to Y$ (proper)

Def. $\mathcal{F} \in (k_X\text{-mod})$ is **constructible** if $X = \coprod_\lambda X_\lambda$ is stratified so that $\mathcal{F}_{X_\lambda}$ is a local system $\forall \lambda$.

Ex: $f: \mathbb{C} \to \mathbb{C}$ by $z \mapsto z^m$

$f_* \underline{k}_{\mathbb{C}} = ?$ For D = disk not thru 0,

$$f_* \underline{k}_{\mathbb{C}}(D) = \underline{k}_{\mathbb{C}}(f^{-1}(D)) \cong k^m$$

For D = disk thru 0, $f_* k_{\mathbb{C}}(D) \cong k$.

$f_* \underline{k}_{\mathbb{C}}|_{\mathbb{C}^*} \cong \mathcal{L}_{(z \mapsto z^m \text{ on } \mathbb{C})}$ & $f_* \underline{k}_{\mathbb{C}}|_0 \cong k$



$\Rightarrow$ $k = f_* \underline{k}_{\mathbb{C}}(D) \xrightarrow{\text{res.}} f_* k_{\mathbb{C}}(D') = k^{\oplus m}$

is invts. under monodromy rep.

This example is more general than it seems!

If $f: C \to C'$ is a non-constant map btwn smooth curves, locally it is just $z \mapsto z^{e_x}$ for e_x = ramific^n index @ x.

[Ramification pts. are $x \in C'$ where f is not smooth!]

Detour: smooth maps in algebraic geometry.

"Smoothness of a map $f: X \to Y$" $\iff$ "submersion of (pseudo)manifolds"

Def: $\mathrm{Spec}\left(A[x_1,\dots,x_n]/(g_1,\dots,g_r)\right) \to \mathrm{Spec}(A)$ is smooth if $\left(\frac{\partial g_i}{\partial x_j}\right)_{\substack{1\le i\le r \\ 1\le j\le n}}$ is of rank r $\forall x$.

i.e. $f: X \to Y$ is smooth iff it induces surjections on tangent spaces.

non-ex: $\pi: \mathrm{Spec}\, k[u,v] \to \mathrm{Spec}\, k[x,y]$ by $x \mapsto u$, $y \mapsto uv$.

$k[u,v] \cong k[x,y][v]/(xv-y)$ [x] vanishes @ $x=0$.

Thm: $\varphi: X \to Y$ is smooth iff φ is flat, locally of finite present^n and its fibers are smooth.

E.g. • $\varphi: \mathbb{C} \to \mathbb{C}$ by $z \mapsto z^m$

Fiber @ 0 is $\mathbb{C}[x]/(x^m)$ ← non-reduced, so not smooth!

• Elliptic curve ex. is smooth on $\mathbb{A}^1 - \{0,1\}$ but not if we extend to 0 or 1

Recap on abelian categories:

Working Def. An abelian category $\mathcal{A}$ is an additive category (i.e. s.t. $\mathrm{Hom}_{\mathcal{A}}(A,B)$ is an abelian gp + mild conditions.) s.t.

$\forall\, A \xrightarrow{\phi} B$ in $\mathcal{A}$, $\exists\, \mathrm{Ker}(\phi), \mathrm{Coker}(\phi) \in \mathcal{A}$ s.t.

$\mathrm{Coim}(\phi) \longrightarrow \mathrm{Im}(\phi)$ is an isomorphism.

(quotient of A) (sub of B)

Defined via universal property: e.g. $\mathrm{Coker}(\phi) \in \mathcal{A}$ is

$$A \xrightarrow{\phi} B \longrightarrow \mathrm{Coker}(\phi)$$

$$B \xrightarrow{\forall f \text{ s.t. } f\circ\phi = 0} C \quad \text{(} \exists! \text{ factorization through } \mathrm{Coker}(\phi)\text{)}$$

Exs:
- ▷ Mod_A = cat. of A-modules
- ▷ $\mathrm{Sh}(X)$ = cat. of sheaves on X

This condition arises when one is not careful w/decorations one adds: e.g. {filtered vector spaces/k} is not abelian, but has Kernels & Cokernels.

Problem: $F^p k := \begin{cases} k & \text{for } p \geq 1 \\ 0 & \text{for } p < 1 \end{cases}$ & $\widetilde{F}^p k = \begin{cases} k & \text{for } p \geq 0 \\ 0 & \text{for } p < 0 \end{cases}$

$f : k \longrightarrow k$ is filtered, ie. $f(F^p k) \subseteq \widetilde{F}^p k$.

$\mathrm{coim}(f) = k$ w/ F-filtn

$\mathrm{im}(f) = k$ w/ $\widetilde{F}$-filtn

Sources of Constructible Sheaves: Cohomology of proper maps!

Next Goal: develop some homological alg. to define $R^i f_* \mathcal{L}$; $H^i(X, \mathcal{L})$.

Reminders on sheaf cohomology:

- Example of a derived functor: measures discrepancy of a functor from being exact.

Let $F: \mathcal{A} \longrightarrow \mathcal{B}$ be a functor (between abelian cats). F is right exact (resp. left exact) if whenever $0 \to A_1 \to A_2 \to A_3 \to 0$ is s.e.s., we have an exact seq $F(A_1) \to F(A_2) \to F(A_3) \to 0$ (resp. $0 \to F(A_1) \to F(A_2) \to F(A_3)$)

Exs: f_* is left exact; $\Gamma(X, -)$ is left exact

$- \otimes F$ is right exact.

Thm/Def: $F: \mathcal{A} \to \mathcal{B}$ a left exact functor. Then its right derived functors $R^iF: \mathcal{A} \to \mathcal{B}$ are functors s.t.:

- ▷ R^iF is additive & $=0$ for $i<0$.
- ▷ $R^0F = F$
- ▷ Have a les $\forall$ses.
- ▷ functorial as map on ses
- ▷ R^iF is universal with these properties
- ▷ $\nexists \forall A \in \mathcal{A}$, $A \hookrightarrow I$ s.t. $R^iF(I) = 0 \; \forall i>0$

Some examples: ▷ $\Gamma(X, -)$ has right derived functor $H^i(X, -)$

▷ Derived functors of $f_*(-)$ gives $R^if_*(-)$: sheaves whose fibers are coh. of fibers (proper base change!)

★ Thm/Def. does not show how to compute! ★

Practical Guide: To compute $F(A)$, replace A by a complex $A \to C^\bullet$.

For all of our purposes, we want A & $C^\bullet$ to be "the same"

▷ In particular, need $H^i(A) = H^i(C^\bullet)$, but this is not enough!

▷ Rather than give formulas, let's first explore what we mean by 2 complexes being "the same".

The Derived Category:

$\mathcal{A}$: abelian category (e.g. $\mathrm{Loc}(X)$, $\mathrm{Sh}(X)$, etc.)

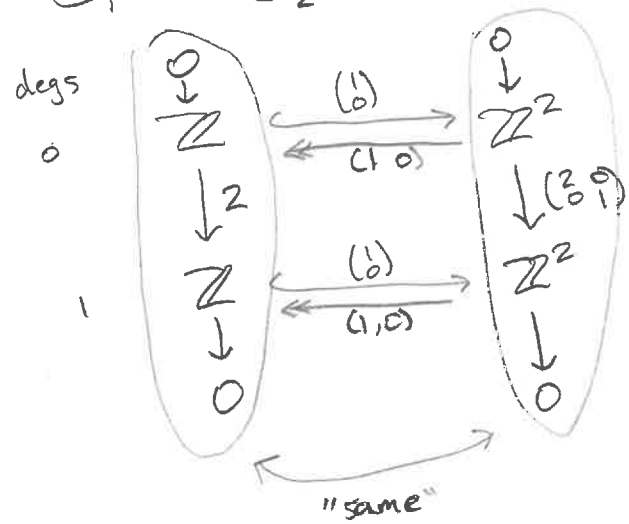
$C(\mathcal{A})$ = category of complexes

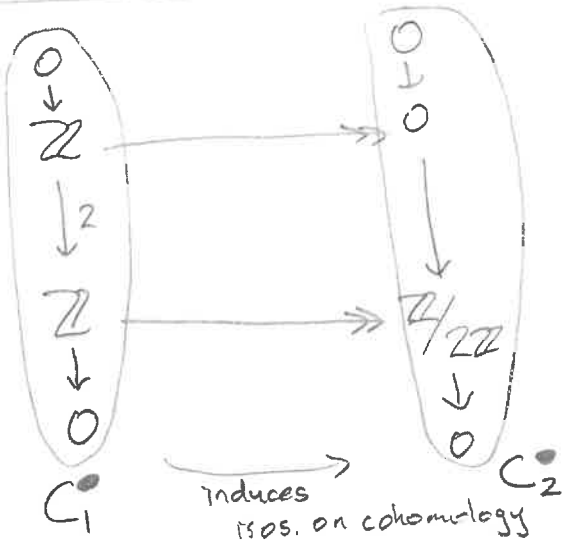
$$\cdots \to A_0 \to A_1 \to A_2 \to A_3 \to \cdots \quad \text{in } \mathcal{A}$$

IDEA: want to treat 2 complexes as "the same" if $C_1^\bullet \to C_2^\bullet$ induces isomorphisms on (co)homology groups

Call these "quasi-isomorphisms"

Ex:





But: there's no inverse!

Working Def: $D(\mathcal{A})$ is a category whose objects are complexes & maps are usual maps + inverses of all quasi-isoms.

e.g. $C_1^\bullet \cong C_2^\bullet$ in $D(\mathcal{A})$.

Cost of this construction is that $D(\mathcal{A})$ is no longer abelian!

$D(A)$ is "triangulated": suppose $0 \to A^\bullet \xrightarrow{f} B^\bullet \xrightarrow{g} C^\bullet \to 0$ is a short exact sequence. For a complex $C^\bullet$, we have a "shift functor" $C^\bullet[1] = C^{\bullet+1}$

$$C^\bullet: \quad \cdots \to C^1 \to C^2 \to C^3 \to \cdots$$

$$C^\bullet[1]: \quad \cdots \to C^2 \to C^3 \to C^4 \to \cdots$$

$$\Longrightarrow \quad H^i(C^\bullet[1]) = H^{i+1}(C^\bullet)$$

Claim/Lem: There is a natural map $C^\bullet \to A^\bullet[1]$ in the derived category.

Pf: Put $\operatorname{cone}(f)^\bullet$:

$$\cdots \to \underset{B^n}{\overset{n}{\overset{A^{n+1}}{\oplus}}} \xrightarrow{\partial} \underset{B^{n+1}}{\overset{n+1}{\overset{A^{n+2}}{\oplus}}} \to \cdots$$

$$\partial(a,b) = (-\partial_A a, \partial_B b - f(a))$$

$$\partial = \begin{pmatrix} -\partial_A & 0 \\ -f & \partial_B \end{pmatrix}$$

$$A^\bullet[1] \xleftarrow[\pi]{\text{proj}^n} \operatorname{cone}(f)^\bullet \xrightarrow{\Psi} C^\bullet$$

$\Psi(a,b) = g(b)$ [well-defined bc $\Psi(\partial(a,b)) = g(\partial_B b - f(a)) = \partial_C g(b)$, with $g \circ f = 0$]

Ψ is a qiso:

$$\operatorname{Ker}(\Psi) = \{(a,b) \mid g(b) = 0\} = A^{n+1} \oplus f(A^n) \cong A^{n+1} \oplus A^n$$

$$\Rightarrow \operatorname{Cone}(1_A)^\bullet: \left[\cdots \to \underset{A^n}{\overset{n}{\overset{A^{n+1}}{\oplus}}} \xrightarrow{\begin{pmatrix} -\partial_A & 0 \\ -1_A & \partial_A \end{pmatrix}} \underset{A^{n+1}}{\overset{n+1}{\overset{A^{n+2}}{\oplus}}} \to \cdots \right]$$

- has trivial cohomology groups
- bc 1_A homotopic to 0: define $A^{n+1} \oplus A^n \xrightarrow{s} A^n \oplus A^{n-1}$, $s(a,a') = (-a', 0)$

Can check $ds + sd = 1_A$.

Reminder on homotopies:

Lem. If $f, g: A^\bullet \to B^\bullet$ are homotopic, i.e.

$$\cdots \to A^{n-1} \xrightarrow{\partial_A} A^n \xrightarrow{\partial_A} A^{n+1} \to \cdots$$
$$\cdots \to B^{n-1} \xrightarrow{\partial_B} B^n \xrightarrow{\partial_B} B^{n+1} \to \cdots$$

with f, g and s as shown, s.t. $f - g = \partial_B s + s \partial_A$

Then f & g induce the same map on cohomology groups

Pf: $f(x) - g(x) = \partial_B(s(x)) + s(\partial_A x) = \partial_B(s(x))$ is a bdry

($x \in \ker(\partial_A)$) $\Rightarrow [f(x)] = [g(x)]$ in $H^n(B^\bullet)$ //

"Triangulated" Structure on $D^b(\mathcal{A})$:

has a collection of "distinguished triangles"

① Every map $A^\bullet \xrightarrow{f} B^\bullet$ fits into a unique distinguished triangle

$$A^\bullet \xrightarrow{f} B^\bullet \xrightarrow{} \mathrm{Cone}(f)^\bullet \xrightarrow{[1]} A^\bullet[1]$$

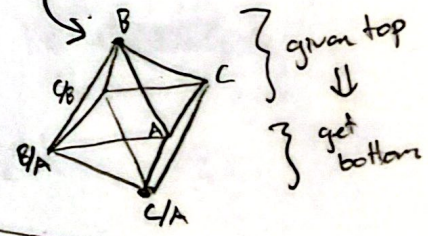
& $\mathrm{Cone}(1_A) \doteq 0$

② $\{$distinguished triangles$\}$ is closed under $[1]$ & rotation $\left(A^\bullet \to B^\bullet \to C^\bullet \xrightarrow{[1]} A^\bullet[1] \rightsquigarrow B^\bullet \to C^\bullet \to A^\bullet[1] \xrightarrow{[1]} B^\bullet[1]\right)$

③
$$\begin{array}{ccccccc} A^\bullet & \to & B^\bullet & \to & C^\bullet & \xrightarrow{[1]} & \\ \downarrow & & \downarrow & & \exists!\downarrow & & \\ A'^\bullet & \to & B'^\bullet & \to & C'^\bullet & \xrightarrow{[1]} & \end{array}$$

Reflection of 3rd isom. thm: $\exists\, C/A$ fitting into $0 \to B/A \to C/A \to C/B \to 0$

④ [Technical] Octahedral Axiom: $\forall\ A \to B \to B/A$ & $\forall\ B \to C \to C/B$



Notes: $\mathrm{Hom}^\bullet$ spaces between cxs. remember

$$\mathrm{Hom}^n(A^\bullet, B^\bullet) = \prod_{p \in \mathbb{Z}} \mathrm{Hom}(A^p, B^{p+n})$$

$\partial^n = \prod_p \partial^n_p$ for $\partial^n_p : \mathrm{Hom}(A^p, B^{p+n}) \to \mathrm{Hom}(A^p, B^{p+n+1})$

by $\partial^n_p(f_p) = \partial^{p+n}_{B^\bullet} \circ f_p + (-1)^{n+1} f_{p+1} \circ \partial^p_{A^\bullet}$

Lem: $\Gamma(X, \mathrm{Hom}^\bullet(F^\bullet, G^\bullet)) = [F^\bullet, G^\bullet] = \{$homotopies $F^\bullet \to G^\bullet\}$

(see Back!)

Pf that $H^0\Gamma(X, \mathcal{H}om^\bullet(A^\bullet, B^\bullet)) = [A^\bullet, B^\bullet]$:

$$\Gamma(X, \mathcal{H}om^n(A^\bullet, B^\bullet)) = \prod_p \mathrm{Hom}_{\mathcal{O}_X}(A^p, B^{p+n})$$

$$\mathrm{Ker}(\partial^0) = \left\{ (f_p) \in \prod_p \mathrm{Hom}_{\mathcal{O}_X}(A^p, B^p) \;\middle|\; \overbrace{\partial_B \circ f - (-1)^n f \circ \partial_A}^{\partial(f_p)} = 0 \right\}$$

↑ chain map cond^n

$$\mathrm{Im}(\partial^{-1}) = \left\{ \partial_B^{p-1} \circ h^p + h^{p+1} \circ \partial_A^p \right\} = \left\{ \text{homotopies} \right\} //$$

$h_p : A^p \to B^{p-1}$

W

Pf. octahedral identity holds?

- Back to derived functors: -

Recall that we want to build "derived functors" $Rf_*, H^*, -\overset{L}{\otimes}F, -$

By: ① Replace F w/ a "good" complex $C^\bullet$ qiso to F.
② Compute $F(C^\bullet)$ & take cohomology.

Idea: replace F w/ qiso. complex $C^\bullet$ s.t. $R^iF(C^a) = 0 \ \forall i > 0$.
→ this depends on F!

ie. To compute $F(\mathcal{F})$ suffices to take a resolution of $\mathcal{F}$ by objects that are F-acyclic, e.g. C is F-acyclic iff any surj^n $B \twoheadrightarrow C \implies F(B) \twoheadrightarrow F(C)$.

Most common "F-acyclics" are projective & injective sheaves:

Projective:

$$\begin{array}{ccccc} & & P & & \\ & \exists\swarrow & \downarrow & & \\ \forall\ A & \twoheadrightarrow & B & \to & 0 \end{array}$$

Injective:

$$\begin{array}{ccccc} 0 & \to & A & \hookrightarrow & B \\ & & \downarrow & \swarrow & \\ & & I & & \end{array}$$

Warning: ▷ $Sh(X)$ has enough injectives, but not enough projectives

▷ Injectives are awful: they never compute anything...

→ $\mathcal{F} \hookrightarrow \prod_{x \in X} i_{x,*} I_x$ for I_x : injective module w/ $I_x \twoheadrightarrow \mathcal{F}_x$.

Injective modules: ▷ over $\mathbb{Z}$, must be divisible: $\mathbb{Q}$ (true for any PID)

▷ If R is a domain & not a field, every injective R-module is ∞-dim'l.

Warning: "Projective" sheaf $\not\Rightarrow$ locally free !
(locally a projective module)

Why? $\mathcal{F} = \mathcal{O}$, $0 \to \mathcal{O}(-2) \to \mathcal{O}(-1)^{\oplus 2} \to \mathcal{O} \to \mathcal{O}$

Γ is not exact on this:

les: $0 \to 0 \to 0 \to k \to k^2 \to k \to 0$

Scanned with CamScanner

What sheaves are acyclic?

▷ Injectives are acyclic $\forall$ left exact F.
↳ standard def^n/construction uses these.

▷ Flasque/flabby: when restriction maps are always surjective

↳ Prop. $\mathcal{F}$: flasque $\Rightarrow$ $\mathcal{F}$ is Γ & f_* acyclic.

Pf: $0 \to \mathcal{F} \xrightarrow{\alpha} \mathcal{G} \xrightarrow{\beta} \mathcal{H} \to 0$ s.e.s. Then $\mathcal{G} \to \mathcal{H}$ is locally surjective $\Rightarrow \exists\, g_i \in \Gamma(U_i, \mathcal{G}) \mapsto h|_{U_i}$

on $U_i \cap U_j$, $\beta(g_i - g_j) = \beta(g_i) - \beta(g_j) = h|_{U_i \cap U_j} - h|_{U_i \cap U_j} = 0 \Rightarrow \exists\, f_{ij} \in \Gamma(U_i \cap U_j, \mathcal{F})$ s.t. $f_{ij} = g_i - g_j$

Extend $f_{ij} \in \Gamma(U_j, \mathcal{F})$ lift & $g_j \mapsto g_j + \tilde{f}_{ij}$.

Now $g_i = g_j$ on $U_i \cap U_j$. Then continue... $\Rightarrow H^1(X, \mathcal{F}) = 0$

Then use LES for $0 \to \mathcal{F} \to \mathcal{I} \to \mathcal{I}/\mathcal{F} \to 0$

▷ For left derived functors, use flat sheaves

$\mathcal{B}^\bullet \longrightarrow \mathcal{F}$: qfso.

NOTE: It is still very hard to compute $f_*\mathcal{F}$ in general! [will eventually do so w/ dec. thm!]

General Philosophy: Better to work w/ complexes than cohomology.

▷ Consider $R^i f_* \mathcal{L}$ in more detail: What is this sheaf roughly? ($\mathcal{L}$ = local system)

other page ① Proper base change: $(R^i f_* \mathcal{L})_y = H^i(X_y, \mathcal{L}|_{X_y})$ ↳ local system on fiber.

② If $\mathcal{L} = \underline{\mathbb{Q}}_X$, Then $H^i(X, \underline{\mathbb{Q}}_X) \cong H^i(X, \mathbb{Q})$

Pf: Let $C^n(X, \mathbb{Q}) =: C^n(X)$ denote singular co-chains & $C^n : [U \mapsto C^n(U)]^\#$. then $\underline{\mathbb{Q}}_X \longrightarrow C^\bullet$ is a resolution. (uses local contractability)

C^n is flabby: can always extend a cochain on U by 0.

$\Gamma(X, C^n) = C^n(X)$, so this proves it. //

③ If not, $H^i(X, \mathcal{L}) =$ cohomology w/ $\mathcal{L}$-coeffs [give formulat example]

other page

Rmk: $R^i f_* \mathcal{L}$ is constructible for $\mathcal{L}$: constructible on X & $f: X \to Y$ proper. (Uses same proof as before!)

$\rightsquigarrow$ **Def.** $\star$ $D_c(X)$ [or $D_c^b(X)$] denotes the full sub-cat. of $D(X) = D(Sh(X))$ [or $D_c^b(X) = D^b(Sh(X))$] of $C^\bullet$ s.t. $H^i(C^\bullet)$ is constructible $\forall i$.

(assume also all local systems involved are finite rank!)

$\star$ $Constr(X) :=$ category of constructible shvs on X.

Prop: $D^b(X) \simeq D^b(Constr(X))$

Pf Idea: for $\mathcal{L} \in Constr(X)$ & $\mathcal{F} \in Sh(X)$, s.t. $\mathcal{F} \to \mathcal{L}$, $\exists\, \mathcal{L}' \to \mathcal{F}$ s.t. $\mathcal{L}' \to \mathcal{F} \to \mathcal{L}$ is surj^n. & $\mathcal{L}' \in Constr(X)$

Build $\mathcal{L}'$ by lifting local gens of $\mathcal{L}$ to $\mathcal{F}$

Philosophy: complexes are better than cohomology.

Exs: ① $R(F \circ G) \cong RF \circ RG$ for left exact F, G

[vs. spectral sequence $R^pF(R^qG) \Rightarrow R^{p+q}(F \circ G)$]

② Universal coeffs: $C^\bullet$ = cx computing $H^i(X, \mathbb{Z})$.

cx computing $H^i(X; A)$ is $C^\bullet \otimes^{\mathbb{L}}_{\mathbb{Z}} A$.

(vs. $H^i(X; A) \cong (H^i(X, \mathbb{Z}) \otimes_{\mathbb{Z}} A) \oplus Tor^1(H^{i+1}(X, \mathbb{Z}), A)$)

Consequence: Often later try to work directly in $D_c^b(X)$.

Next Goals: ▷ 6 functor formalism in $D_c^b(X)$ [& introduction to $f_!, f^!$.

▷ Verdier duality in $D_c^b(X)$

$\rightsquigarrow$ opens up calculus of the derived category!

f_* & f^* are adjoint functors:

Prop. $f: X \to Y$. This is a canonical isom. (functorial in each input) $\mathrm{Hom}(f^*\mathcal{F}, \mathcal{G}) \cong \mathrm{Hom}(\mathcal{F}, f_*\mathcal{G})$

Rmk: In coherent category, this is locally:

X $\quad$ $B\text{-}\mathrm{GM}^G$

$\downarrow$ $\quad$ $\uparrow$

Y $\quad$ $A\text{-}\mathrm{GM}^F$

$\mathrm{Hom}_B(M \otimes_A B, N) \cong \mathrm{Hom}_A(M, N)$

Special Case of:

$\mathrm{Hom}_B(M \otimes_A P, N) \cong \mathrm{Hom}_A(M, \mathrm{Hom}_B(P, N))$

Pf: $\phi \in$ RHS, $\mathcal{F} \to f_*\mathcal{G}$.

From $\forall V \supset f^*(U)$, $f^{-1}(V) \supset U$

$\Longrightarrow$ $\mathcal{F}(V) \to f_*\mathcal{G}(V) = \mathcal{G}(f^{-1}V) \xrightarrow{\text{res}} \mathcal{G}(U)$

gives $\varinjlim_{V \supset f^{-1}(U)} \mathcal{F}(V) \to \mathcal{G}(U)$.

In reverse: $\Psi: f^*\mathcal{F} \to \mathcal{G}$.

$\forall V \subset Y$ open, $f(f^{-1}V) \subset V$ gives

$$\mathcal{F}(V) \to \varinjlim_{W \supset f(f^{-1}(V))} \mathcal{F}(W) = f^*\mathcal{F}(f^{-1}(V)) \to \mathcal{G}(f^{-1}(V)) = f_*\mathcal{G}(V).$$

Can check these are inverse operations on stalks //

Consequences of an adjoint pair (f^*, f_*):

- f^* is right exact; f_* left exact [true for abelian cats.]
- More generally, left adj. preserves colimits: cokernels, direct limits, stalk
  right adj. preserves limits: kernels, products, pullback, inverse lims, ...

OVER →

6 functor formalism on $D^b_c(X)$: [a framework across many different situations!]

Def of $f_!$: For $s \in \Gamma(U, F)$, $\operatorname{supp}(s) = \{x \in X \mid s(x) \neq 0\}$.

e.g. If $U = \operatorname{Spec}(A)$ & $F = \widetilde{M}$ is coherent, $s \in M$ has

$$\operatorname{supp}(s) = \{\mathfrak{p} \in \operatorname{Spec}(A) \mid s \neq 0 \text{ in } M_{\mathfrak{p}}\}$$
$$= \{\mathfrak{p} \mid \nexists a \in A \setminus \mathfrak{p} \text{ s.t. } a \cdot s = 0\} \cong \operatorname{Spec}(A/\operatorname{ann}_A(s))$$
$$\rotatebox{90}{$\subseteq$}$$
$$\mathfrak{p} \supset \operatorname{ann}_A(s)$$

▷ $\Gamma_c(U, F) \subseteq \Gamma(U, F)$ is set of sections with compact support.

▷ Γ_c remains left exact & so has a right derived functor $R\Gamma_c$ [when $F = \underline{A}_X$, computes $H^i_c(X, A)$.]

$$R\Gamma_c \xrightarrow{\text{coh.}} H^i_c(X, F^\bullet) \text{ or } H^i_c(X, F).$$

▷ $f_! F$ is given by:

$$f_! F(U) = \left\{ s \in \Gamma(f^{-1}(U), F) \;\middle|\; f|_{\operatorname{supp}(s)} : \operatorname{supp}(s) \to U \text{ is proper} \right\}$$

supports are relatively cpt.

$f_! F \hookrightarrow f_* F$; isom. for proper F.

▷ $Rf_! F$ can be computed w/ "c-soft" shvs, i.e. those for which $\Gamma(X, F) \xrightarrow{\text{res}} \Gamma(K, F)$ is surjective $\forall$ compact $K \subset X$

[Here, $\Gamma(K, F) := \varinjlim_{U \supset K} \Gamma(U, F)$]

▷ $f_!$ & $Rf_!$ satisfy base change w/o properness assumptions!

Thm (Base Change).

$$\begin{array}{ccc} X' & \xrightarrow{g} & X \\ {\scriptstyle f'}\downarrow & \lrcorner & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g'} & Y \end{array}$$

Cartesian. Then

▷ $(g')^* \circ f_! = f'_! \circ g^*$ in $\operatorname{Sh}(Y')$

▷ $(g')^* \circ Rf_! = Rf'_! \circ g^*$ in $D^b(Y')$

e.g. $j : U \hookrightarrow X$ open
$j_! F$ = extension by zero

/ **e.g.** $\mathbb{A}^2 - \{(0,0)\} \hookrightarrow \mathbb{A}^2$
$j_! \underline{\mathbb{Q}}$ = ext by zero; $(j_! \underline{\mathbb{Q}})_0 = 0$
$i_* \underline{\mathbb{Q}} = \underline{\mathbb{Q}}$; $(j_! \underline{\mathbb{Q}})_0 = \mathbb{Q}$

Def of $f^!$: —Very mysterious functor in general—

Hope: maybe $f_!$ has an adjoint functor $f^!$, i.e.,

$$\mathrm{Hom}(f_! \mathcal{F}, \mathcal{G}) \cong f_* \mathcal{H}om(\mathcal{F}, f^! \mathcal{G})$$

NOTE: $A = (A_n)_{n \in \mathbb{Z}}$ on $\mathbb{Z} = \{\dots \cdot \cdot \cdot \dots\} \xrightarrow{f} \{pt\}$

$$f_* A = \prod_{n \in \mathbb{Z}} A_n \quad , \quad f_! A = \bigoplus_{n \in \mathbb{Z}} A_n$$

Having a left adjoint $\overset{\text{in abelian cat}}{\Longrightarrow}$ left exact

$\Longrightarrow$ preserves limits

$f_!$ cannot have left adjoint bc

$$\bigoplus_n \prod_k A \neq \prod_k \bigoplus_n A$$

→ Cannot exist in $Sh(X)$ also bc $f_!$ is not right exact

But maybe it exists in $D^b_c(X)$:

① $j : U \hookrightarrow X$: open inclusion

$$\Gamma(U, j^! \mathcal{G}) = \mathrm{Hom}_U(\underline{A}_U, j^! \mathcal{G}) = \Gamma(U, \mathcal{H}om(\underline{A}_U, j^! \mathcal{G}))$$

$$= \Gamma(U, \mathcal{H}om(j_! \underline{A}_U, \mathcal{G})) = \mathrm{Hom}(j^* j_! \underline{A}_U, j^* \mathcal{G})$$

$$= \mathrm{Hom}(\underline{A}_U, j^* \mathcal{G}) = \Gamma(U, j^* \mathcal{G})$$

$\Longrightarrow j^! = j^*$ for open inclusions.

② $j : U \hookrightarrow X$; $f : X \to Y$

$$\Gamma(U, f^! \mathcal{G}) = \mathrm{Hom}_U(\underline{A}_U, j^* f^! \mathcal{G}) = \mathrm{Hom}_X(j_! \underline{A}_U, f^! \mathcal{G})$$

$$= \Gamma(Y, f_* \mathcal{H}om(j_! \underline{A}_U, f^! \mathcal{G}))$$

$$= \Gamma(Y, \mathcal{H}om(f_! j_! \underline{A}_U, \mathcal{G}))$$

$$= \mathrm{Hom}_Y(f_! j_! \underline{A}_U, \mathcal{G})$$

$\Longrightarrow$ $f^! \mathcal{G}$ should be $[U \longmapsto \mathrm{Hom}(f_! j_! \underline{A}_U, \mathcal{G})]$.

NOT a sheaf! → replace $\underline{A}_X$ with a resol^n $\underline{A}_X \to K^\bullet$ [by c-soft sheaves]

$\Longrightarrow$ Can now define $f^!$ on derived category!

Method for $f^! \mathcal{G}$:

① $\mathcal{G} \longrightarrow \mathcal{I}^\bullet$ an injective resolution.

$\underline{A}_X \longrightarrow \mathcal{K}^\bullet$: flat c-soft res^{n}.

If $\mathcal{I}$ is injective, $f^!_{\mathcal{K}} \mathcal{I}(U) = \mathrm{Hom}_Y(f_! \mathcal{K}_U, \mathcal{I})$

$\mathcal{K} : U \mapsto f_!(\mathcal{K}^\bullet|_U)$ is a "cosheaf" valued in $Sh(Y)$

& $f^! \mathcal{G} := \mathrm{Hom}_{D(X)}(\mathcal{K}, \mathcal{G})$

Simple Cases:

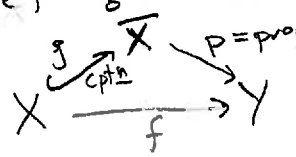
Ⓐ If $j: U \hookrightarrow X$ is open, then $j^! \cong j^*$ [exact] & $j^!$ is defined on sheaves &

Ⓑ If $j: Z \hookrightarrow X$ is c.s.d, then $j^!$ is defined on sheaves &

$$j^! \mathcal{G}(U \cap Z) = \Gamma_{U \cap Z}(U, \mathcal{G})$$

(supp contained in $U \cap Z$)

Ⓒ Better def^{n}:



$\Rightarrow f_! = p_* \circ j_!$

Prop. $f^! = j^* \circ p^!$ (reduces to p: proper.)

for projective $X \xrightarrow{f} Y$ also have $f^! = i^! \pi^!$

$X \xrightarrow{i} \mathbb{P}^n_Y \xrightarrow{\pi}$ projection

i: clsd

$\mathbb{P}^n_Y = Y \times \mathbb{P}^n$

▷ $i^!$ is done from before

▷ π is controlled by "dualizing ox": $\pi^! \mathcal{G} \cong \mathcal{G} \boxtimes \omega_{\mathbb{P}^n}$

$\omega_{\mathbb{P}^n} = \underline{A}_{\mathbb{P}^n}[2n]$

Ⓓ $f: X \to Y$ smooth:

$$f^! \mathcal{F} = f^* \mathcal{F} \otimes \omega_{X/Y}[d], \quad \omega_X = p_X^! \underline{A}_{pt}$$

constr. setting: $\omega_{X/Y} = or_{X/Y} = or_X \otimes_{\underline{A}} f^*(or_Y)^\vee \overset{\text{if } X, Y \text{ are smooth}}{=} \underline{A}_X[2d]$ (d = rel.dim)

$\omega_X \otimes f^* \omega_Y^\vee$

Check for a local chart $U \cong \mathbb{R}^n \to \{pt\}$ using adjunction.

étale: $f^! \mathcal{F} = f^* \mathcal{F}(d)[2d]$.

[redacted]

Cor: $H^m(i_x^! \underline{\mathbb{Q}}_X) \cong \varinjlim_{U \ni x} H^m(U, U - x; \mathbb{Q})$

[vs. $H^m(i_x^* \underline{\mathbb{Q}}_X) \cong \begin{cases} \mathbb{Q}, & m=0 \\ 0, & m>0 \end{cases}$ always.]

Exact Δ: $U \xrightarrow{j} X \xleftarrow{i} X \setminus U$

$$Rj_! \circ j^!(\mathcal{F}) \longrightarrow \mathcal{F} \longrightarrow Ri_* \circ i^*(\mathcal{F}) \xrightarrow{[1]}$$

(unit & counit of adjunctions)

why? On stalks: $(j_! j^! \mathcal{F})_x = (j_! j^* \mathcal{F})_x = \begin{cases} \mathcal{F}_x & \text{if } x \in U \\ 0 & \text{if } x \notin U \end{cases}$

$$(i_* i^* \mathcal{F})_x = \begin{cases} \mathcal{F}_x & \text{if } x \in \partial X \setminus U \\ 0 & \text{if } x \in U \end{cases}.$$

Verdier Duality $\Longrightarrow$ Poincaré Duality for smooth X:

① Universal Coeffs: (A: Dedekind domain) $\forall$ open $U \subset X$, $\exists$ split ses

$$0 \to \mathrm{Ext}\left(\mathbb{H}_c^{q+1}(U, F^\bullet), A\right) \to \mathbb{H}^{-q}(U; \mathbb{D}_X F^\bullet) \to \mathrm{Hom}\left(\mathbb{H}_c^q(U, F^\bullet), A\right) \to 0$$

$f: X \to pt$, $Rf_* R\mathcal{H}om(\mathbb{D}_X F^\bullet, \mathbb{D}_{pt}(Rf_! F^\bullet)) \cong \mathbb{D}_{pt}(Rf_! F^\bullet)$

Why? $R\Gamma(U, \mathbb{D}_X F^\bullet) \cong R\mathrm{Hom}_A(C^\bullet, A)$ for $C^\bullet = R\Gamma_c(U, F^\bullet)$

Then compute $R\mathrm{Hom}_A(C^\bullet, A)$ w/ spectral seq: A is dedek. dom. $\Rightarrow \exists\ A \to I^0 \to I^1$ inj. resol.

In particular, if A = field,

$$\mathbb{H}^{-q}(U, \mathbb{D}_X F^\bullet) \cong \mathrm{Hom}\left(\mathbb{H}_c^q(U, F^\bullet), A\right)$$

Prop. (Poincaré Duality). X (A: field) smooth of real dim n. Then

$$H^{n-i}(X, \mathcal{L}^\vee \otimes \mathcal{O}_X) \cong H_c^i(X, \mathcal{L})^\vee.$$

Pf: $\mathbb{D}_X \mathcal{L} \cong \mathcal{L}^\vee \otimes \omega_X \cong \mathcal{L}^\vee \otimes \mathcal{O}_X[n]$

Then $\mathbb{H}^{-i}(X, \mathbb{D}_X \mathcal{L}) \cong \mathbb{H}^{n-i}(X, \mathcal{L}^\vee \otimes \mathcal{O}_X) = H^{n-i}(X, \mathcal{L}^\vee \otimes \mathcal{O}_X)$

$\cong$ // univ. coeffs. $\cong$ P.D.

$\mathrm{Hom}(H_c^i(X, \mathcal{L}), A) = H_c^i(X, \mathcal{L})^*$ //

Define: $f = i : Y \hookrightarrow X$ clsd inclusion of sm. mfds.

$$\mathbb{H}^*(X, X-Y; F^\bullet) := \mathbb{H}^*(X, Ri_! i^! F^\bullet)$$ [equal to usual notion for $F^\bullet = \underline{A}_X$]

Prop. (Alexander Duality): X oriented, $\dim X = n$, $i : Y \hookrightarrow X$ clsd, A = field.

$$\Longrightarrow \quad H^{n-i}(X, X-Y; A) \cong H_c^i(Y, A)^*.$$

Pf: $i_! i^! \mathbb{D}_X = i_! \mathbb{D}_Y i^* = \mathbb{D}_X i_! i^*$ (using $i_! = i_*$). Apply to $\underline{A}_X \mapsto (\mathbb{D}_X \underline{A}_X)[-n]$

$$\Longrightarrow \quad i_! i^! \underline{A}_X[n] = \mathbb{D}_X i_! \underline{A}_Y. \quad \text{Apply } \mathbb{H}^{-k}(X, -).$$

$$H^{n-k}(X, X-Y; A) \cong \mathbb{H}^{n-k}(X; Ri_! i^! \underline{A}_X) = \mathbb{H}^{-k}(X, \mathbb{D}_X i_! \underline{A}_Y)$$

$$\cong H_c^k(X, i_! \underline{A}_Y)^* \cong H_c^k(Y, \underline{A}_Y)^* = H_c^k(Y, A)^* \quad //$$

(univ. coeffs)

The Intersection Cohomology Complex:

Goal: sheafify intersection homology

So far: if X is **smooth**:

$H^*(X, \mathbb{Q}) \rightsquigarrow \underline{\mathbb{Q}}_X$ s.t. $H^*(X, \underline{\mathbb{Q}}_X) \cong H^*(X, \mathbb{Q})$

Poincaré Duality $\rightsquigarrow$ Verdier duality $\mathbb{D}_X$

~~X: locally contractible; $X = X_n \supset X_{n-1} \supset \cdots \supset X_0 \supset X_{-1} = \emptyset$ Whitney stratified~~

Recall: to compute $H^*(X, \underline{\mathbb{Q}}_X)$, we used a **resol^n** of $\underline{\mathbb{Q}}_X$:

Variant for homology: $C^\bullet \longrightarrow \underline{\mathbb{Q}}_X$

$C^{-i}(\mathcal{U}) = C_i(\mathcal{U})$: i-chains on $\mathcal{U}$

∂ = connecting map for $C_\bullet(\mathcal{U})$.

Generalize to IC: p = perversity fcn,

$IC^{-i}_p(\mathcal{U}) := I^{BM}C^p_i(\mathcal{U}) \rightsquigarrow IC^\bullet_p \in D^b(X)$

Recall: $\mathcal{F} \in Sh(X)$ is **soft** if $\forall$ cpt $K \subset X$,

$\Gamma(X, \mathcal{F}) \xrightarrow{res} \Gamma(K, \mathcal{F}) := \varinjlim_{U \supset K} \Gamma(U, \mathcal{F})$ is surjective

Lem: IC^{-i}_p is a soft sheaf (& so computes $Rf_*, Rf_!$)

$\Longrightarrow$ **Prop:** $I^{BM}H^p_i(X) = \mathbb{H}^{-i}(X, IC^\bullet_p) := R^{-i}\Gamma(IC^\bullet_p)$

$IH^p_i(X) = \mathbb{H}^{-i}_c(X, IC^\bullet_p) := R^{-i}\Gamma_c(IC^\bullet_p)$

In fact $IC^\bullet_p$ will not be enough simple objects for category of perverse shvs later: even on **smooth** X, also have $C^\bullet(\mathcal{L})$ computing $H^i(X, \mathcal{L})$ for $\mathcal{L}$: local system on X

[Return to $H^*(X, \mathcal{L})$ here now...]

Define $IC^\bullet_p(\mathcal{L})$ similarly.

$I^{BM}H^p_i(X, \mathcal{L}) = \mathbb{H}^{-i}(X; IC^\bullet_p(\mathcal{L}))$ & $IH^p_i(X, \mathcal{L}) = \mathbb{H}^{-i}_c(X, IC^\bullet_p(\mathcal{L}))$

(Co)Homology with coefficients in a local system:

for $\sigma : \Delta^i \longrightarrow X$ & $\mathcal{L}$: local system $\Longrightarrow \sigma^*\mathcal{L}$: loc. sys. on Δ^i

Note Δ^i = contractible, so $\sigma^*\mathcal{L}$ = constant shf.

$\mathcal{L}_\sigma := \sigma^*\mathcal{L}(\Delta^i)$ = fiber of $\mathcal{L}|_{\sigma(\Delta^i)}$; $\forall p \in \Delta^i$, get rest^n map

$P_p^\sigma : \mathcal{L}_\sigma \xrightarrow{\sim} \mathcal{L}_{\sigma(p)} = (\sigma^*\mathcal{L})_p$

$p \in \tau \subset \sigma$: $P_\tau^\sigma = (P_p^\tau)^{-1} \circ P_p^\sigma : \mathcal{L}_\sigma \longrightarrow \mathcal{L}_\tau$

$$C_i(X, \mathcal{L}) = \left\{ \sum_\sigma l_\sigma \cdot \sigma \;\middle|\; \begin{array}{l} \Sigma \text{ is finite (modify for BM)} \\ l_\sigma \in \mathcal{L}_\sigma \end{array} \right\}$$

$$\partial_i\left(\sum_\sigma l_\sigma \sigma\right) = \sum_\sigma \sum_{\tau : \text{face of } \sigma} \pm \underbrace{P_\tau^\sigma(l_\sigma)}_{\text{only difference from normal homology.}} \cdot \tau$$

$$H_i(X, \mathcal{L}) = \mathrm{Ker}(\partial_i) / \mathrm{Im}(\partial_{i+1})$$

Ex: $H_i(\mathbb{C}^\times, \mathcal{L}_m) = ?$ [$\mathcal{L}_m$ = local system for monodromy $z \mapsto z^m$.]

e.g. $\mathbb{Z} \longrightarrow \mathrm{Aut}(A) = A^\times$, $n \longmapsto \lambda^n$

$\mathbb{C}^\times \underset{\text{homotopy}}{\sim} S^1$, so suffices to do S^1: $S^1 =$  has triangulation T w/ $\sigma : \Delta^1 \to S^1$.

$C_1^T(\mathbb{C}^\times, \mathcal{L}_m) = C_0^T(\mathbb{C}^\times, \mathcal{L}_m) = A$.

$$0 \longrightarrow A \xrightarrow{\lambda - Id_A} A \longrightarrow 0$$

the isomorphism matters!

$$H_0(\mathbb{C}^\times, \mathcal{L}_m) = \mathrm{Ker}(\zeta_m - 1) = \begin{cases} 0 & \text{if } m \neq 1 \\ k & \text{if } m = 1 \end{cases}$$

$$H_1(\mathbb{C}^\times, \mathcal{L}_m) = k/(\zeta_m - 1) = \begin{cases} 0 & \text{if } m \neq 1 \\ k & \text{if } m = 1 \end{cases}.$$

▷ Can define similarly for IH_i

▷ Have all similar formulas as H_i : MV seqs, [Poincaré duality], if fibers are v.s.: $IH_i(X, \mathcal{L}) \cong IH^i(X, \mathcal{L})^*$

▷ $C_*(X, \mathcal{L}) = C_*(X_H) \underset{G/H}{\otimes} M$ & $C^*(X, \mathcal{L}) = \mathrm{Hom}^*_{G/H}(C_*(X_H), M)$.

$H \subset \mathrm{Ker}(\rho_2)$; M = fiber

▷ $H^0(X; \mathcal{L}) \cong M^{\pi_1(X)}$.

Q: What does $IC_p^\bullet(\mathcal{L})$ look like as a complex of shvs?
e.g. What are its stalk? How does it interact w/ Whitney strata?

...

Prop: $x \in X_{n-k} - X_{n-k-1}$. Then

$$\mathcal{H}^i(IC_p^\bullet(\mathcal{L}))_x \cong \begin{cases} 0 & , i > p_k - n \\ IH^p_{-i-(n-k+1)}(L_x, \mathcal{L}) & , i \leq p_k - n \end{cases} \cong I^{BM}H_i^p(U, \mathcal{L}) \quad U \cong \mathbb{R}^{n-k} \times cL_x$$

Pf: Let $U \ni x$ be a nghd small enough s.t.

$$U \cong \mathbb{R}^{n-k} \times \operatorname{cone}(L_x), \quad \dim(L_x) = k-1$$

$$H^i(IC_p^\bullet(U, \mathcal{L})) \cong H_i(I^{BM}C^p_{-\bullet}(U, \mathcal{L})) \cong H_{i+n-k}(I^{BM}C^p_{-\bullet}(cL_x, \mathcal{L}))$$

$$\cong I^{BM}H^p_{-n+k-i}(cL_x, \mathcal{L}) \underset{\text{cone formula}}{=} \begin{cases} 0 & , -n+k-i < k-1-p_k+1 \\ & \Leftrightarrow p_k - n < i \\ IH^p_{-n+k-i-1} & , p_k - n \geq i \end{cases}$$

//

$$U_k = X - X_{n-k} \underset{\text{open}}{\subset} X$$

$$U_k \xhookrightarrow{j_k} U_{k+1} \xhookleftarrow{i_k} U_{k+1} - U_k = X_{n-k} - X_{n-k-1}$$

Prop: @ $\mathcal{H}^{-i}(Rj_{k*}IC_p^\bullet|_{U_k})_x \cong I^{BM}H_i^p(U \cap U_k; A)$
for U disting. nghd (i.e. $U \cong \mathbb{R}^{n-k} \times cL_x$)

(b) $\mathcal{H}^j(IC_p^\bullet|_{U_{k+1}}) = 0$ if $j > p_k - n$ $\forall k \geq 2$.

(c) $IC_p^\bullet|_{U_{k+1}} \longrightarrow Rj_{k*}j_k^*(IC_p^\bullet|_{U_{k+1}})$ induces isoms.

$$\mathcal{H}^j(i_k^* IC_p^\bullet|_{U_{k+1}}) \xrightarrow{\sim} \mathcal{H}^j(i_k^* Rj_{k*}j_k^* IC_p^\bullet|_{U_{k+1}}) \quad \forall j \leq p_k - n.$$

IC equals its lower-* extension on $U_{k+1} - U_k$ up to degree $p_k - n$.

Lem: $IC_p^\bullet$ is constructible w.r.t. Whitney strat^n on X.

(d) [Pf:

Together these conditions form axioms:

⓪ - normalization ② - vanishing condition

① - lower bound ③ - attaching condition

(actually normal^{n} + lower bd are weaker...)

⓪ $\mathcal{A}^\bullet|_{U_2} \cong \mathcal{L}[n]$ (fixed $\mathcal{L}$: local system)

① $\mathcal{H}^j(\mathcal{A}^\bullet) = 0 \quad \forall j < -n$

② $\mathcal{H}^j(\mathcal{A}^\bullet|_{U_{k+1}}) = 0 \quad \forall j > p_k - n$

③ $\mathcal{H}^j(i_k^* \mathcal{A}^\bullet|_{U_{k+1}}) \cong \mathcal{H}^j(i_k^* R j_{k*} j_k^* \mathcal{A}^\bullet|_{U_{k+1}}) \quad \forall j \leq p_k - n$

These axioms will end up characterizing $IC_p^\bullet$!

→ Deligne's Complex: Define $\mathcal{G}_p^\bullet$ inductively by:

- On U_2, $\mathcal{G}_p^\bullet|_{U_2} = \underline{A}_{U_2}[n]$
- If we know $\mathcal{G}_p^\bullet|_{U_k}$, we put

$$\mathcal{G}_p^\bullet|_{U_{k+1}} = \underbrace{\tau_{\leq p_k - n}}_{\text{truncates cx into degs. } \leq p_k - n} R j_{k*} \mathcal{G}_p^\bullet|_{U_k}$$

Thm: If $\mathcal{A}^\bullet \in D^b(X)$ satisfies axioms ⓪–③, then $\mathcal{A}^\bullet \underset{\text{glob.}}{\cong} \mathcal{G}_p^\bullet$

Pf: Induct on U_k. True on U_2 trivially. Given $\mathcal{A}^\bullet|_{U_k} \cong \mathcal{G}_p^\bullet|_{U_k}$,

$$\mathcal{A}^\bullet|_{U_{k+1}} \overset{②}{\cong} \tau_{\leq p_k - n} \mathcal{A}^\bullet|_{U_{k+1}} \overset{\text{adjunction}}{\underset{⊛}{\longrightarrow}} \tau_{\leq p_k - n} R j_{k*} j_k^* (\mathcal{A}^\bullet|_{U_{k+1}})$$

$$\cong \tau_{\leq p_k - n} R j_{k*} (\mathcal{A}^\bullet|_{U_k}) \cong \tau_{\leq p_k - n} R j_{k*} (\mathcal{G}_p^\bullet|_{U_k}) =: \mathcal{G}_p^\bullet|_{U_{k+1}}.$$

⊛ is iso. bc we can apply ③ to truncation using ②

Cor. ⓪ + ③ $\Longrightarrow$ constructible.

$$IC_p^\bullet \cong \mathcal{G}_p^\bullet$$

Rmk:
- All works equally well w/ $\mathcal{L}$ in place of $\underline{A}_{U_2}$. (maybe just do this!)
- Note $A_{U_2}[n] = \mathbb{O}_{U_2}$ [or $\mathcal{L} \otimes \mathbb{O}_{U_2}$]

A different formulation of axiom (3):

$$\tau_{\le}i^!\to id\to Rj_*j^*\xrightarrow{[1]}\xrightarrow{i^*} i^!\to i^*\to i^*Rj_*j^*\xrightarrow{[1]}$$

$i=i_k$, $j=j_k$: (3) $\iff \mathcal{H}^i(i_k^!\mathcal{A}^\bullet|_{U_{k+1}})=0 \quad \forall i\le p_k-n+1$

For $x\in U_{k+1}\setminus U_k = X_{n-k}\setminus X_{n-k-1}$ & $f_x:\{x\}\hookrightarrow \underbrace{X_{n-k}-X_{n-k-1}}_{\text{sm mfd of dim } n-k}$,

$f_x^!$ computed by $f_x^!\cong f_x^*(-)[-(n-k)]$.

(bc compute cpt coh. of a nghd of x in $X_{n-k}-X_{n-k-1}$.)

$$i_x^!\mathcal{A}^\bullet\cong f_x^*(i_k^!\mathcal{A}^\bullet)[-n+k]$$

So phrasing (3) in terms of stalks: ($\overbrace{\ }$ = shift from $f_x^!$)

(3') "Co-stalk vanishing": $\mathcal{H}^i(i_x^!\mathcal{A}^\bullet)=0 \quad \forall i<\overbrace{p_k-k+1}^{p_k-n+1+\overbrace{n-k}}$ & $\forall x\in X_{n-k}-X_{n-k-1}$

$\mathcal{H}^i(i_x^*\mathcal{A}^\bullet)\cong \mathbb{H}^i(U_x,\mathcal{A}^\bullet)$

$\mathcal{H}^i(i_x^!\mathcal{A}^\bullet)\cong \mathbb{H}_c^i(U_x,\mathcal{A}^\bullet)$

Consequences for IH hold for IC:

- ▷ Kunneth: $IC^\bullet_{X_1\times X_2}\cong IC^\bullet_{X_1}\boxtimes IC_{X_2}$
- ▷ X: cx variety $\Rightarrow IC^\bullet_m\cong IC^\bullet_n$ (call this $IC^\bullet_X$)

Thm (P.D.): A: field. $p+q=t$ (complementary perversities).

Then $\exists$ qiso. $D_X(IC^\bullet_q)[n]\cong IC^\bullet_p$

Pf: need to show $D_X IC^\bullet_q[n]$ satisfies axioms for $IC^\bullet_p$.

(0) $j:U_2\hookrightarrow X$: $j^*D_X IC^\bullet_q[n]\cong D_{U_2}j^!IC^\bullet_q[n]\cong D_{U_2}(\underline{A}_{U_2}[n])[n]\cong D_{U_2}(\underline{A}_{U_2})$

$= \omega_{U_2}\cong \underline{A}_{U_2}[n]$.

(2) $x\in X_{n-k}-X_{n-k-1}$. $\mathcal{H}^j(i_x^!IC^\bullet_q)=0 \quad \forall i\le q_k-k+1$ by co-stalk vanishing.

so: $\mathcal{H}^j(D_X IC^\bullet_q[n])_x\cong \mathcal{H}^{j+n}(D_X IC^\bullet_q)_x\cong \mathbb{H}^{j+n}(U_x, D_X IC^\bullet_q)$

$\overset{UCT}{\cong} \mathbb{H}_c^{-j-n}(U_x,IC^\bullet_q)^*\cong \mathcal{H}^{-j-n}(i_x^!IC^\bullet_q)^*=0$ for

$-j-n\le q_k+k+1 \iff j\ge k-n-q_k-1 = p_k-n+1$

($q_k+p_k=k-2$)

(3) sim. $\mathcal{H}^j(i_x^!D_X IC^\bullet_q[n])\cong \mathcal{H}^{j+n}(i_x^!D_X IC^\bullet_q)=\mathbb{H}_c^{j+n}(U_x,D_X IC^\bullet_q)\overset{UCT}{\cong}\mathbb{H}^{j-n}(U_x,IC^\bullet_q)^*$

$\cong \mathcal{H}^{-j-i}(i_x^*IC^\bullet_q)^*=\mathcal{H}^{-j-i}(IC^\bullet_q)_x^*=0$ for $-j-n>q_k-n\leftrightarrow j\le p_k-k+1$

① Same as ②. //

Cor: ∃ nondeg. pairing

$$I^{BM}H_i^p(X,A) \otimes I H_{n-i}^q(X,A) \longrightarrow A .$$

Pf:
$$I^{BM}H_i^p(X,A) \cong \mathbb{H}^{-i}(X, IC_p^\bullet) \cong \mathbb{H}^{-i}(X, D_X IC_q^\bullet[n])$$
$$\cong \mathbb{H}^{n-i}(X, D_X IC_q^\bullet) \overset{UCT}{\cong} \mathbb{H}_c^{-n+i}(X, IC_q^\bullet)^* \cong I H_{n-i}^q(X,A)^* \;//$$

Small adaptation gives:

$$D_X(IC_q^\bullet(\mathcal{L}^\vee)[n]) \cong IC_p^\bullet(\mathcal{L})$$

& so also get:

$$I^{BM}H_i^p(X;\mathcal{L}^\vee) \cong I H_{n-i}^q(X,\mathcal{L})$$

Q: All of this still depends on a Whitney stratif^n.
Is $IC_p^\bullet$ independent of choice?

Thm: $\mathcal{X}_1, \mathcal{X}_2$: 2 Whitney strat^n's on X. Then

$$IC_{p,\mathcal{X}_1}^\bullet \underset{\text{qiso.}}{\simeq} IC_{p,\mathcal{X}_2}^\bullet .$$

Pf Sketch: ▷ Find equivaleat, intrinsic axioms that are equiv. to ours & show they characterizes $IC_p^\bullet$ indep. of strat^n.

Intrinsic Axioms: $\mathcal{A}^\bullet \in D(X)$:

$$\operatorname{supp}^j(\mathcal{A}^\bullet) = \overline{\{x \in X \mid \mathcal{H}^j(i_x^* \mathcal{A}^\bullet) \neq 0\}}$$
$$\operatorname{cosupp}^j(\mathcal{A}^\bullet) = \overline{\{x \in X \mid \mathcal{H}^j(i_x^! \mathcal{A}^\bullet) \neq 0\}}$$
$$p^{-1}(j) = \begin{cases} \min\{k : p(k) \geq j\} & , j \leq p(n) \\ \infty & , j > p(n) \end{cases}$$

⓪ (normal^n) $\exists\, U \subset X$ open s.t. $\mathcal{A}^\bullet|_U \cong \omega_U = \underline{A}_U[n]$ } same
① (lower bd) $\mathcal{H}^j(\mathcal{A}^\bullet) = 0 \quad \forall j < -n$
②′ (support) $\dim \operatorname{supp}^j(\mathcal{A}^\bullet) \leq n - p^{-1}(j+n) \quad \forall j > -n$
③ (cosupport) $\dim \operatorname{cosupp}^j(\mathcal{A}^\bullet) \leq n - q^{-1}(j) \quad \forall j < 0$. for q: compl. perversity

on back →

▷ We have already shown $IC^{\bullet}_{p,\mathcal{X}}$ satisfies these intrinsic axioms.

▷ In opposite direction, choose a strat^{n} $\mathcal{X}: X_n \supset X_{n-1} \supset \cdots \supset X_0$ s.t. stalk & costalk coh. are locally constant on each strata.

▷ Let $S_k = X_{n-k} - X_{n-k-1}$ codim k. $x \in S_k$:

if $H^r(i_x^* A^\bullet) \neq 0$, the support bound $\Rightarrow$ $-k \leq n - p^{-1}(r) \iff r \leq p(k) - n$

$$\Rightarrow H^r(i_x^* A^\bullet) = 0 \quad \forall r > p_k - n$$

& cosupport $\longrightarrow$ $H^r(i_x^! A^\bullet) = 0 \quad \forall r < -q_k = p_k - n + 2$

$$\underbrace{i^! A^\bullet}_{\text{in degs } r \geq -q_k = p_k - n + 2} \longrightarrow \underbrace{i^* A^\bullet}_{r \leq p_k - n} \longrightarrow i^* R j_* j^* A^\bullet \xrightarrow{[1]} \cdots$$

$\Rightarrow H^r(i^* A^\bullet) \longrightarrow H^r(i^* R j_* j^* A^\bullet)$ is isom. $\forall r \leq p_k - n$.

and $H^r(i^* A^\bullet) = 0 \quad \forall r > p_k - n$

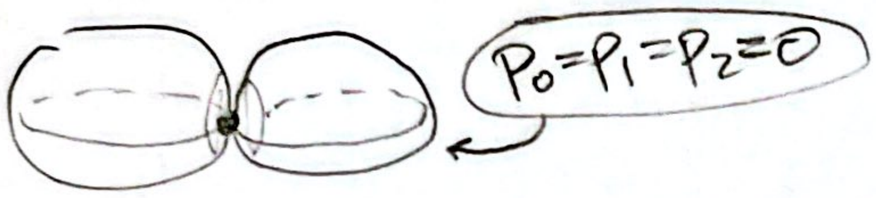
$\Rightarrow i^* A^\bullet \cong \tau_{\leq p_k - n}(i^* R j_* j^* A^\bullet)$ [Deligne's construction.]

Example:

$\mathbb{P}^1 \cup \mathbb{P}^1 \subset \mathbb{P}^2 \Rightarrow$ $j: \mathbb{A}^1 \sqcup \mathbb{A}^1 \hookrightarrow \mathbb{P}^1 \cup \mathbb{P}^1$

$IC^\bullet = \tau_{\leq p_2 - n} Rj_* \underline{A}_{U_2}[2]$

$= \tau_{\leq(-2)} Rj_* \underline{A}_{U_2}[2]$.



$\mathcal{H}^r\left(i_x^* R j_* \underline{A}_{U_2}[2]\right) = \varinjlim_{V \ni x} H^{r+2}(V \setminus \{x\}, A)$ ($S^1 \sqcup S^1$)

$= \begin{cases} A^2 & , r = -2, -1 \\ 0 & , \text{else} \end{cases}$

$\Rightarrow \mathcal{H}^r(i_x^* IC^\bullet) = \begin{cases} A^2 & , r = -2 \\ 0 & , \text{else} \end{cases}$

In fact: $IC^\bullet \cong \nu_* \underline{A}_{\mathbb{P}^1 \sqcup \mathbb{P}^1}[2]$

$\nu: \mathbb{P}^1 \sqcup \mathbb{P}^1 \to \mathbb{P}^1 \cup \mathbb{P}^1$ normal^n

computed via exact Δ

$\underline{A}_X \to \nu_* \underline{A}_{\tilde{X}} \to A_x$ (also computes ω_X.)

Rmks: (common confusion on shifts)

$\mathcal{H}^i(F^\bullet[j]) = \mathcal{H}^{i+j}(F^\bullet)$ moves coh. more negative $\Rightarrow i$

($F^\bullet[j]$ is shifted left/negative direction!)

Relation with normaliz^n:

Prop: $\nu: \tilde{X} \to X$: normaliz^n. Then $IC^\bullet_{X,p} \cong R\nu_* IC^\bullet_{\tilde{X},p}$

Pf: Check axioms: $\nu = \mathrm{Id}$: on smooth locus $\Rightarrow$ normaliz^n.

ν: proper $\Rightarrow$ by proper base change, $i_x^* R\nu_* IC_{\tilde{X}}^\bullet \cong \bigoplus_{y \in \nu^{-1}(x)} i_y^* IC_{\tilde{X}}^\bullet$

$\Rightarrow \{x : \mathcal{H}^r(i_x^* R\nu_* IC_{\tilde{X}}^\bullet) \neq 0\} = \nu\left(\{y : \mathcal{H}^r(i_y^* R\nu_* IC(\tilde{X})^\bullet) \neq 0\}\right)$

preserves dims $\Rightarrow$ support condition

Proper base change for costalks gives same for co-support.

Rmk: Works fine w/local system coeffs: $\mathcal{L}$: loc. sys. on $X^{sm} = U$.

$\Rightarrow IC^\bullet_{X,p}(\mathcal{L}) \cong R\nu_* IC^\bullet_{\tilde{X},p}(\nu^* \mathcal{L}|_{\nu^{-1}(U)})$

Ex: Rational Homology Mfds: (or $\mathbb{Q}$-mfd)

Def: X is a rational homology mfd. of dim m if

$$\forall x \in X,\quad H_i(X, X-x; \mathbb{Q}) = \begin{cases} \mathbb{Q}, & i=m \\ 0, & \text{else.} \end{cases}$$

Ex: If M is a smooth mfd & G: finite gp, $G \curvearrowright M$. Then

$$X = M/G \text{ is a } \mathbb{Q}\text{-mfd.}$$

Prop. X is a $\mathbb{Q}$-mfd $\iff$ the link of any $x \in X$ is a rational homology sphere, ie.

$$H_i(L_x, \mathbb{Q}) = \begin{cases} \mathbb{Q}, & i = \dim L_x \\ 0, & \text{else.} \end{cases}$$

Pf: use excision to write

$$H_i(X, X-x, \mathbb{Q}) \cong H_i(U_x, U_x - x, \mathbb{Q}) \quad \text{for disting. nghd } U_x \text{ \&}$$

then use cone computation. //

Thm. ($p = m$ middle perversity.) X: cx alg. var. of dim n which is a $\mathbb{Q}$-mfd. Then $IC^\bullet_m \cong \mathbb{Q}_X[2n]$.

Pf: $IC_X := IC^\bullet_m[-n]$ (in degs $-n, \dots, n$ now)

▷ normaliz^n & bdd-ness are obvious

▷ $\mathcal{H}^j(\underline{\mathbb{Q}}_X[n])_x \cong \mathbb{H}^j(U_x; \underline{\mathbb{Q}}_X[n]) = H^{n+j}(U_x, \mathbb{Q}) = 0$

when $j+n > 0$ since U_x is contractible.

(so holds for $j \geq -n + \text{codim}$)

▷ $x \in X_{n-2k}$; $\dim(L_x) = 2k-1$ & L_x is $\mathbb{Q}$-homology sphere

(Σ even bc cx alg. variety.)

$$\forall x \in X_{n-2k},\quad \mathcal{H}^j(i_x^! \underline{\mathbb{Q}}_X[n]) \cong \mathbb{H}^j_c(U_x; \mathbb{Q}_X[n]) = H^{j+n}_c(U_x; \mathbb{Q})$$

$$\underset{\text{Kunneth}}{\cong} \bigoplus_{p+q=j+n} H^p_c(\mathbb{C}^{n-k}; \mathbb{Q}) \otimes H^q_c(cL_x, \mathbb{Q})$$

$$= \begin{cases} 0, & j \neq n \\ \mathbb{Q}, & j = n \end{cases} \implies \mathcal{H}^j(i_x^! \mathbb{Q}_X[n]) = 0$$

$\forall\, j \leq n - \text{codim} = n - 2k.$